

AI-Based Parental Control Router: KidSafeNet

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Outline:



Design Methodology

Implementation

Result

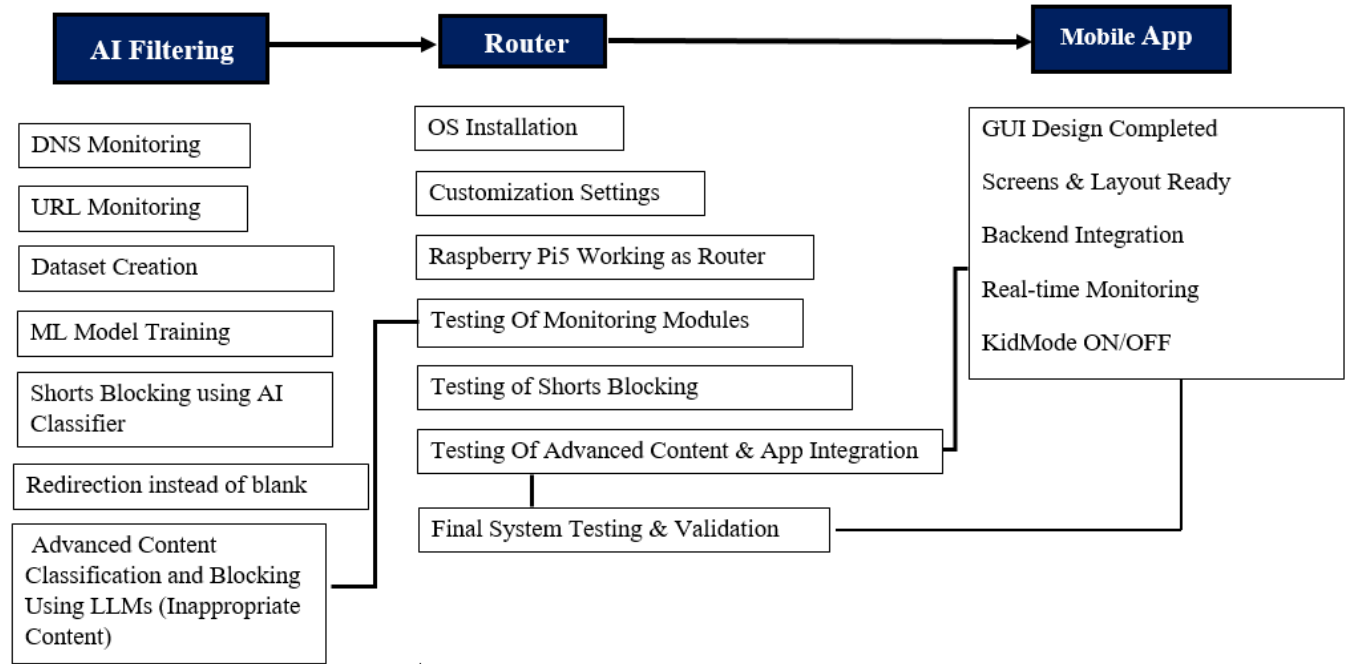
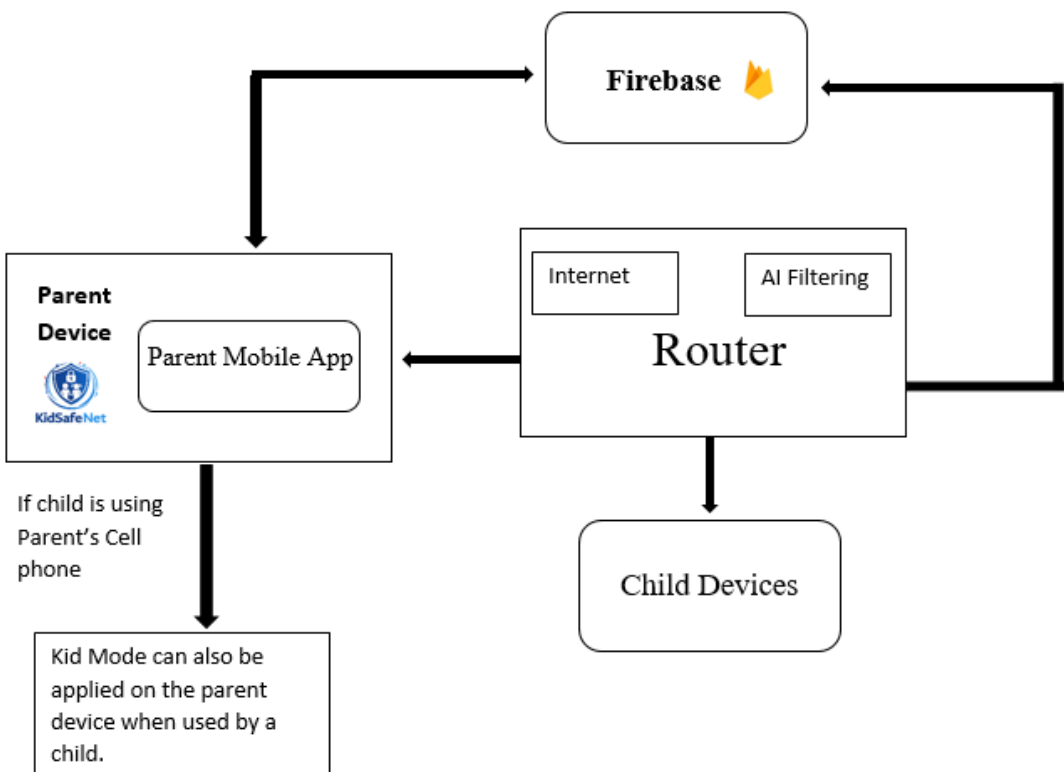
Conclusion

Future Work



KidSafeNet

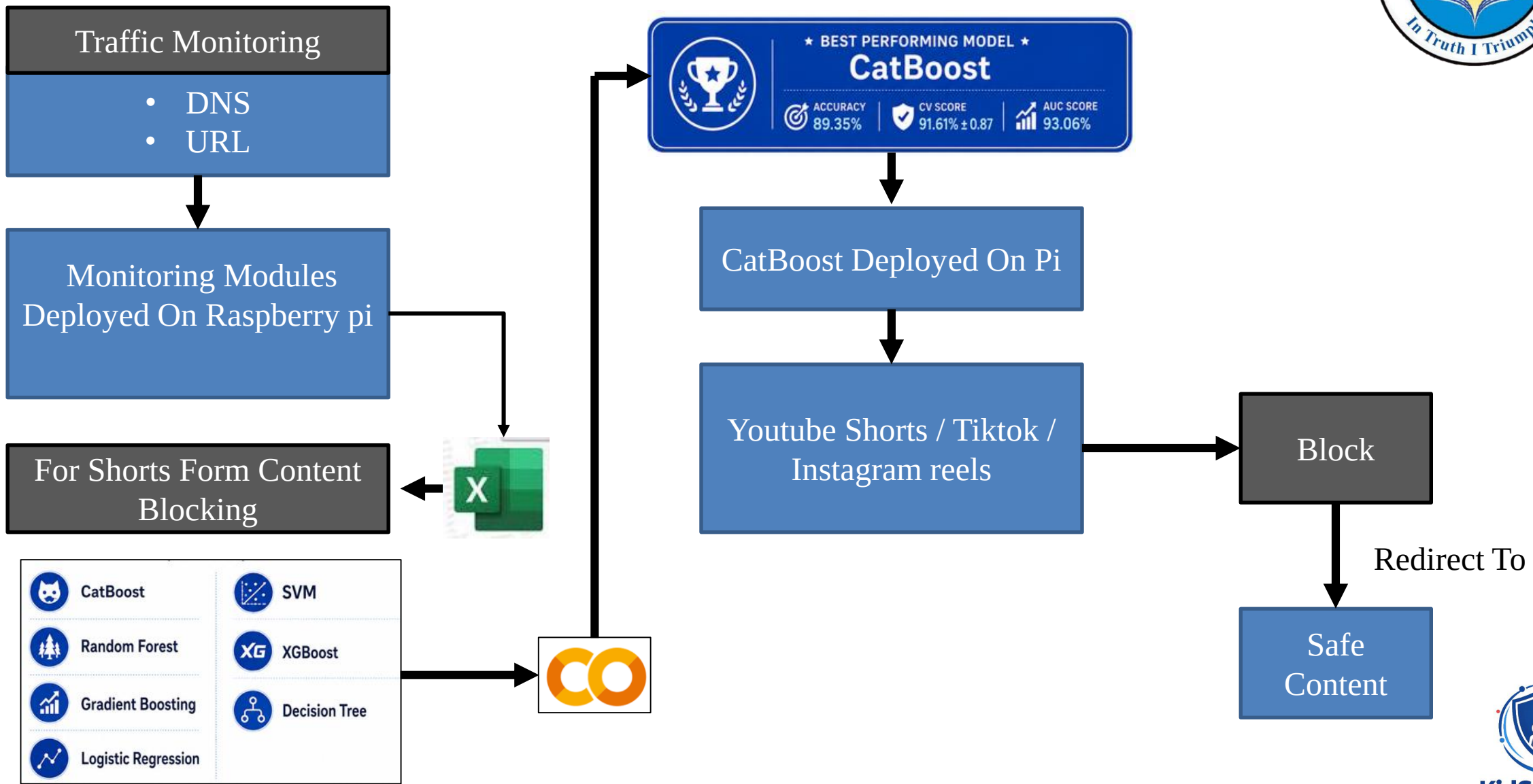
Design Methodology [1/4]



AI Based Parental Control Router



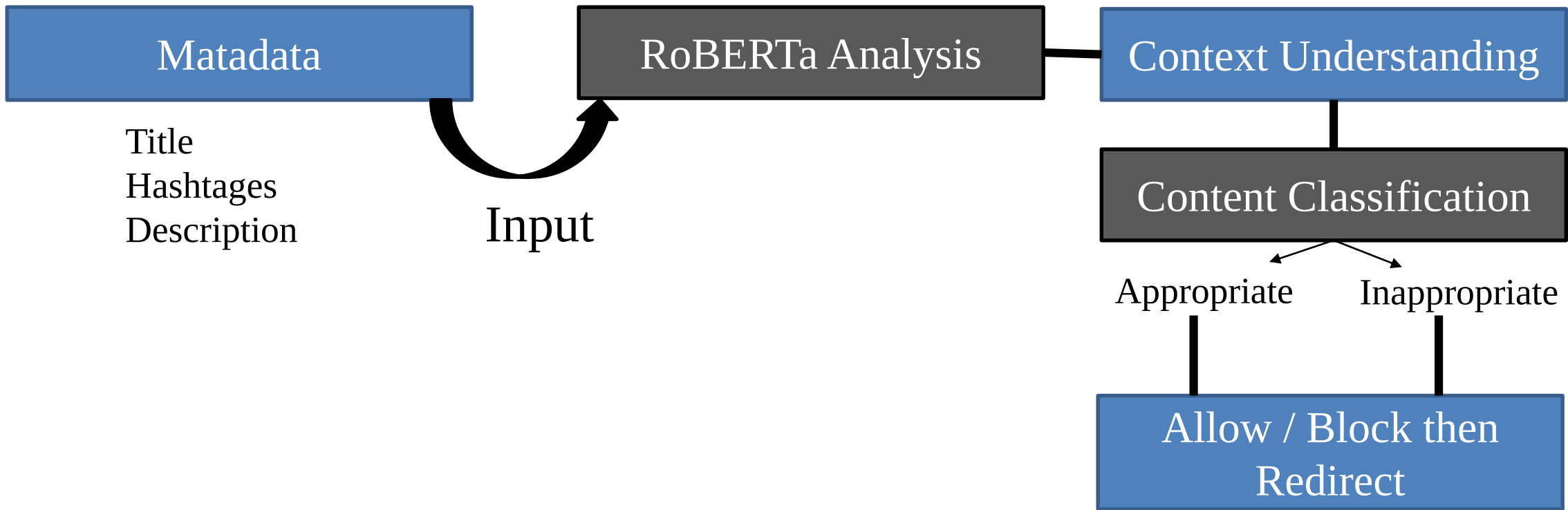
Design Methodology [2/4]



Design Methodology [3/4]



Advanced Context-Aware Filtering Using RoBERTa



This provides intelligent contextual filtering instead of simple keyword matching.



Design Methodology [4/4]



Customized Safe Content Environment

The customized environment:

Removes Inappropriate recommendations.

Filters inappropriate thumbnails.

Hides addictive Shorts content.

Promotes educational and child-friendly content.

Kid Mode ON

AI filtering, blocking, and redirection enabled.

Inappropriate Search Protection

The search request is analyzed.

Inappropriate searches are blocked.

The user is redirected to a safe Content.

Kid Mode OFF

Normal internet access with no filtering.



KidSafeNet

Implementation



HARDWARE COMPONENTS



Raspberry Pi 5

Acts as the smart router and processing unit.



64GB MicroSD Card

Stores OS, applications and system data.



Ethernet Cable

Provides stable internet connectivity.



Raspberry Pi Cooling Case

Keeps the Raspberry Pi cool for stable performance.



27W Power Supply

Provides reliable power to Raspberry Pi 5.

SOFTWARE COMPONENTS



Raspberry Pi OS

Operating system used to run the router and all services.



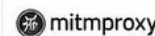
Python

Used for traffic monitoring, data processing and system modules.



Scapy

For DNS packet capture and live network monitoring.



MITM Proxy

Intercepts and decrypts HTTPS traffic to extract complete URLs.



CatBoost

ML model used for traffic classification and content detection.



RoBERTa

Semantic analysis of titles, descriptions, hashtags and metadata.



Flutter

Used to develop the mobile application for parental control.



Firebase

Provides authentication, real-time database and cloud synchronization.



Internet



Raspberry Pi 5
(Router)



Wi-Fi
Hotspot



Child Devices



Parent Mobile App
(Monitoring)



AI Filtering &
Parental Control



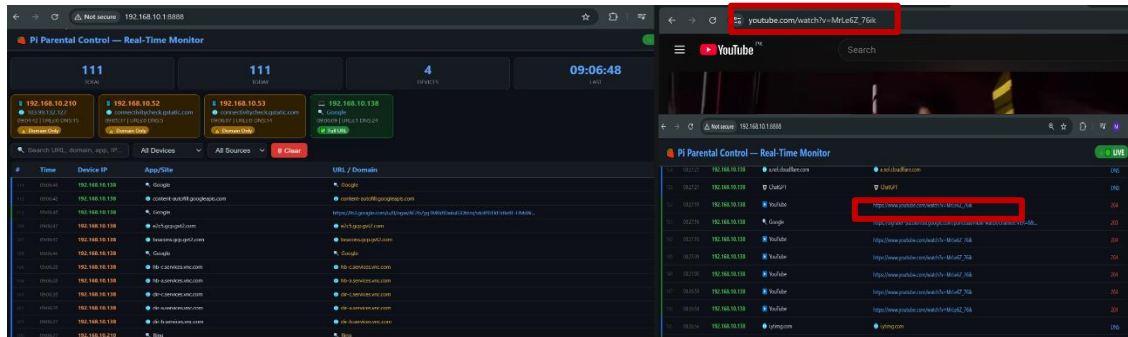
KidSafeNet

Result [1/2]:

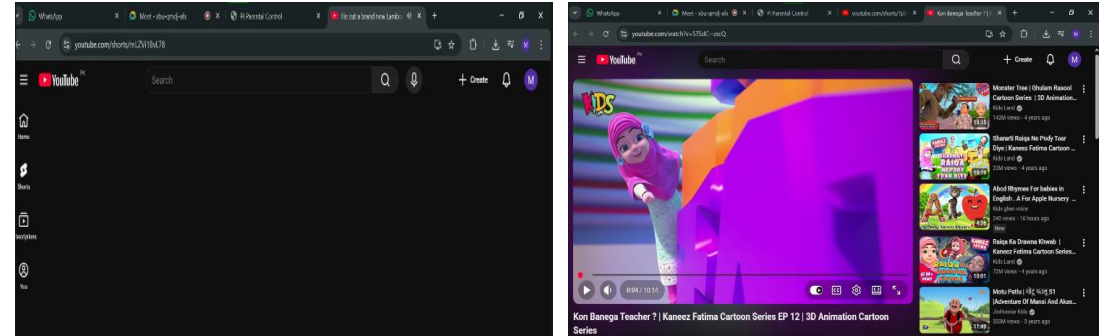
1 Raspberry Pi Working As Router



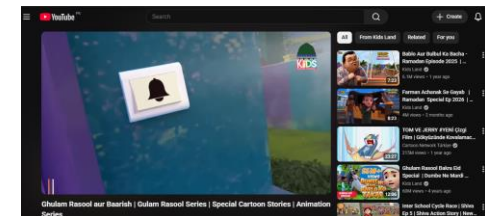
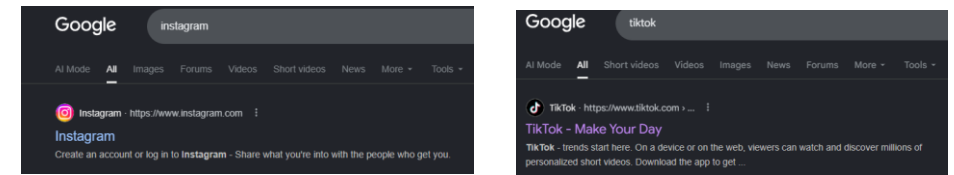
2 Monitoring Modules On Raspberry Pi 5



3 Shorts Blocking and Redirection



4 Instagram & Tiktok Blockage and redirection



5

7

AI filtering is ACTIVE on all connected devices

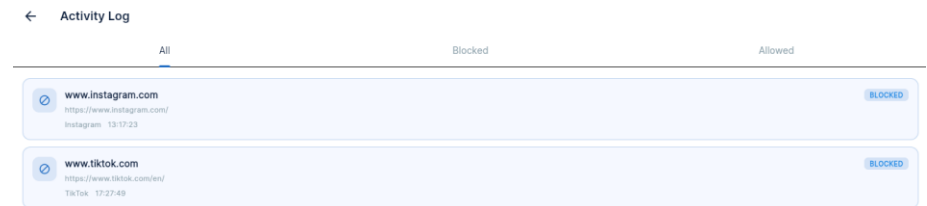
ON



6

Tap to enable AI-based content filtering

OF



Sustainable Development Goals (SDGs):



SDG 16 Peace, Justice & Strong Institutions

- Protects children in digital environments
- Encourages ethical and transparent monitoring



KidSafeNet

Distribution of Work:



Noor –Ul- Aen

AI Filtering

- Traffic Monitoring
- Dataset Generation & ML Model Training
- AI Filtering & Logic

noortec59@gmail.com



Abdul Shakoor

Router Setup

- Raspberry Pi Setup
- AI Model Deployment
- Real-time Blocking System

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Omair Amir

Mobile App Development

- Flutter UI Design
- App Screens & Dashboard
- User Interface Flow

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KidSafeNet

Future Work

Develop a standalone mobile application without requiring a router.

Implement age-based personalized filtering and child profiles.

Enhance AI models for more accurate content understanding and classification.



KidSafeNet

Conclusion:



Developed KidSafeNet, an AI-based parental control router for safer internet usage.

Successfully monitored and analyzed encrypted internet traffic in real time.

Implemented AI-based filtering to detect inappropriate and addictive content.

Applied smart blocking and safe redirection mechanisms.

Provided centralized parental control through a Flutter-Firebase mobile application.



KidSafeNet



Thank You For
Listening
Any Query ?

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KidSafeNet



Backup Slides



KidSafeNet



Logistic Regression

- A simple linear classification model
- Used to predict probability of a class

How it works:

- Takes input features and multiplies them with weights
- Adds them together to get a score
- Applies a sigmoid function to convert it into probability
- If probability $> 0.5 \rightarrow$ BLOCK, else $\rightarrow$ ALLOW



SVM (Support Vector Machine)

- A model that separates data using a boundary (line or curve)

How it works:

- Finds the best boundary that separates two classes
- Keeps maximum distance between classes (called margin)
- Uses closest points (support vectors) to define boundary
- RBF kernel allows curved boundary for complex data



Decision Tree

- A rule-based model (like a flowchart)

How it works:

- Splits data based on conditions (if-else)
- Chooses the best feature at each step
- Continues splitting until final decision is reached
- Final output comes from leaf nodes



Random Forest

- A group of multiple decision trees

How it works:

- Creates many decision trees using random data samples
- Each tree gives its own prediction
- Final result is based on majority voting



Gradient Boosting

- A model that improves step by step

How it works:

- First model makes prediction
- Finds errors in prediction
- Next model focuses on correcting those errors
- Repeats this process to improve accuracy



XGBoost

- An advanced and optimized boosting algorithm

How it works:

- Works like gradient boosting
- Builds models step by step to fix errors
- Uses regularization to avoid overfitting
- Faster and more efficient



Implementation [5/6]:

CatBoost (Best Model)

A powerful boosting algorithm designed for structured data

How it works:

Builds trees sequentially like boosting

Learns from previous mistakes

Handles sparse and complex data efficiently

Reduces overfitting automatically



Introduction:

Children today spend major time online both learning and entertainment.

Short-form videos (YouTube Shorts, Reels, TikTok) are **highly addictive**.

Studies link these platforms to **reduced focus and short-term memory** in children.

Parents need **intelligent, simple tools** to create a healthy digital environment.

Our project introduces an **AI-based router** that provides **smart AI Filtering based parental control**.

Problem Statement:



Existing parental control systems rely on manual keyword or category filters.

These methods cannot reliably detect short-form videos or automatically adapt to new content patterns.

Most tools operate on a single device, not across the whole home network.

Existing systems are signature-based and do not use AI.

Goal: Develop an **AI-powered parental control router** with AI based Filtering.



Literature Review/Background:

There are many limitations in existing solutions

- Software-Based Parental Control Apps (that are just able to block or allow Domains not content and Not AI Based)
- Router-Based Parental Control Systems (that are keyword-Based no AI)

No existing solution combines Network-level monitoring With AI.

Literature Review [Existing Tools]:



System	Type	Key Features	Limitations
Google Family Link [1]	Software(Mobile App)	Screen-time limits, app blocking, location tracking, Mobile lock	No AI ; cannot detect short-form videos, No Router
Net Nanny [2]	Software(Mobile App)	Keyword and category-based filtering	Static, rule-based; No AI; No Router, cannot detect short-form videos
Bark [3]	Software(Mobile App)	Basic AI for text and chat analysis (cyberbullying, self-harm detection)	AI limited to text only; cannot analyze video/images; no Router, No network level filtering
TP-Link Deco (Homecare) [4]	Hardware + (Mobile App)	Router-level filtering only when rules will set from app, Trend Micro web reputation updates, Rule based Filtering	No AI; static categorization; cannot block short-form content
Circle Home Plus [5]	Hardware + (Mobile App)	Time limits, app filters, network monitoring, Router-level filtering only when rules will set from app	No AI; rule-based filtering; no short-video detection; costly subscription

Literature Review[Paper Related Literature]:

S. No.	Year of Publication	Title	Methodology	Limitations
1[6]	2021	Parental Control with Edge Computing and 5G Networks	Proposed a privacy-preserving parental control protocol using edge computing and AI in 5G networks. The system introduces “Smart Response” instead of simple blocking, and uses Private Set Intersection (PSI) and Private Set Operations (PSO) protocols for privacy. The method relies on AI classifiers deployed at the edge to detect harmful content and provide automatic, network-level parental control while preserving both child and parent privacy.	Focused only on text and website content; audio, video, and image analysis are left for future work. Relies on 5G infrastructure support and assumes cooperation from network operators. Implementation is limited to simulation and cryptographic performance, without full real-world deployment.

Literature Review[Paper Related Literature]:



S. No.	Year of Publication	Title	Methodology	Limitations
2[7]	2025	To Block Or Not To Block? Evaluating Parental Controls Across Routers, DNS Services, and Software	Conducted a comparative analysis of seven parental control systems (three routers, two DNS resolvers, one software tool). Tested each by visiting one million domains (Tranco dataset) and recording blocked sites. Domains were categorized (e.g., Adult, Gambling, Hate, Terrorism) using Cisco Umbrella data. Evaluated blocking consistency, overblocking, and underblocking through network measurements and packet captures.	Found that systems frequently fail to block up to 96% of sensitive content and sometimes overblock legitimate websites. Filtering was mostly DNS-based and inconsistent across devices. Some solutions acted as black boxes, lacking transparency and context-aware AI, making their operation unreliable.

Literature Review[Paper Related Literature]:



S. No.	Year of Publication	Title	Methodology	Limitations
3[8]	2024	TikGuard: A Deep Learning Transformer-Based Solution for Detecting Unsuitable TikTok Content for Kids	The study proposes TikGuard, a transformer-based deep learning approach for classifying TikTok videos into categories (Safe, Adult, Harmful, Suicide). It fine-tunes TimesFormer, ViViT, and VideoMAE models using a curated dataset named TikHarm. Key steps include video frame extraction, normalization, data augmentation, and model fine-tuning with optimized hyperparameters. Performance was evaluated using accuracy, precision, recall, and F1-score with TimesFormer achieving the best accuracy ($\approx 86.7\%$).	Limited dataset size (3,948 videos) restricts model generalization. The dataset focuses only on visual features, excluding audio and textual content that could provide better context. Lacks multimodal fusion and cross-platform testing. Comparison with existing systems is limited due to dataset uniqueness.

Literature Review[Paper Related Literature]:



S. No.	Year of Publication	Title	Methodology	Limitations
4[9] (Base Paper)	June 2025	A Router-Based Parental Control Tool for Safe Social Network Usage	Router-based parental control using deep packet inspection (DPI) for analyzing and managing social network traffic	Prototype only; lacks AI-driven decision-making; it works purely on rule-based packet inspection.
5[10]	2024	Age-Based Content Controlling System Using AI for Children	Deep Learning (CNN) for detecting user age and controlling inappropriate media content	Achieved 98.83% accuracy; still experimental; not tested on live online data



Progress Presentation 1

Design Methodology [2/17]



Module 1

AI Filtering

DNS Monitoring ✓

URL Monitoring ✓

AI Content Classification ✗

Module 2

Router

OS Installation ✓

Customization Settings ✓

Raspberry Pi5 Working as Router ✓

Testing Of Monitoring Modules ✗

AI Filtering & App Integration ✗

Final System Testing & Validation ✗



Module 3

Mobile App

GUI Design Completed ✓

Screens & Layout Ready ✓

Backend Integration Pending ✗

Real-time Monitoring Pending ✗

Design Methodology/ Module 1 [3/17]



The module1 is divided into **two major steps**

Both steps are designed to **move toward the final AI-based goal**

Focus is on:

- **Network-level monitoring**
- **Application Level Monitoring (URL Monitoring)**

These steps build the **foundation for future AI integration**

Two Main Phases

DNS (Domain Name System) Based Network Traffic Monitoring

URL (Uniform Resource Locator) Monitoring using MITM Proxy

Design Methodology Module 1 [4/17]



DNS-Based Network Traffic Monitoring

Implemented **DNS-based network traffic monitoring**

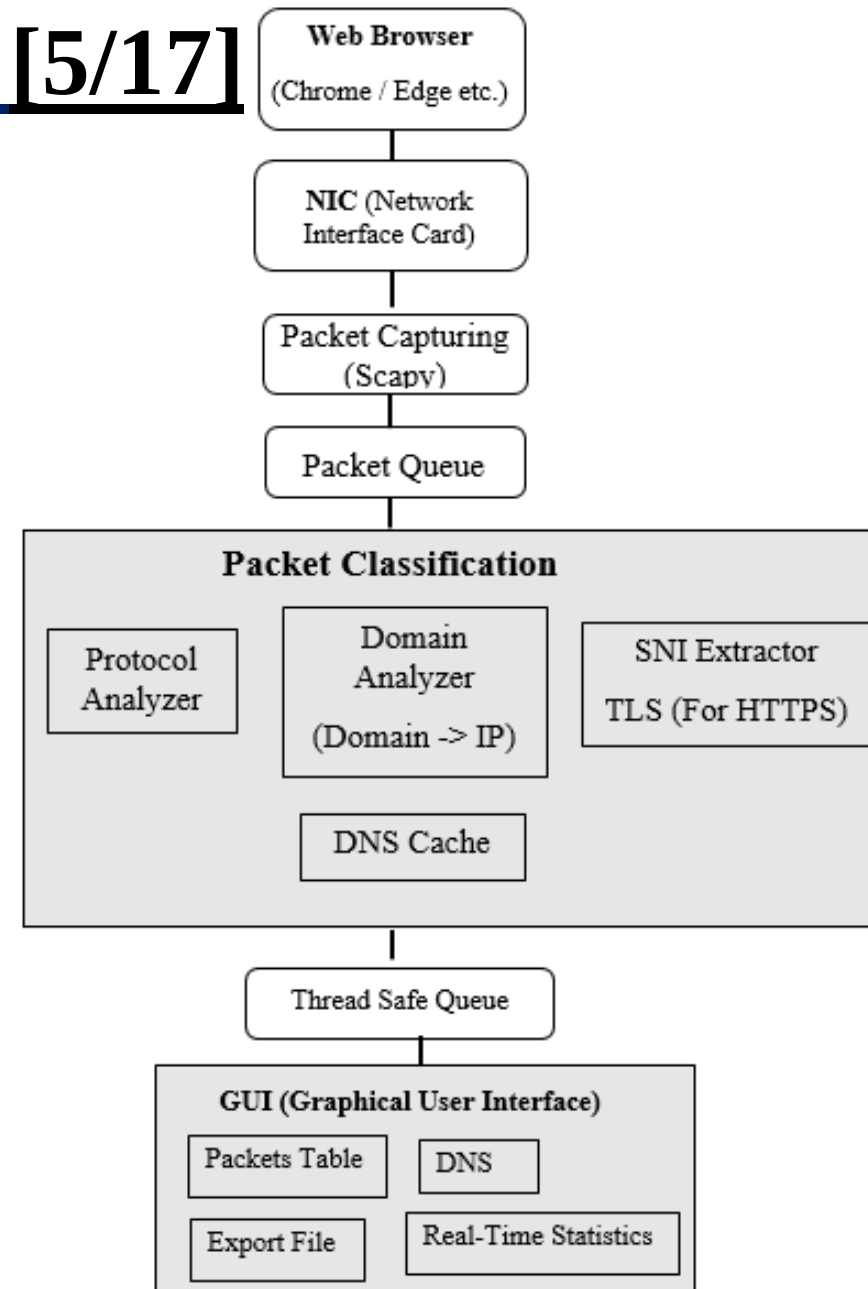
Captured live packets using **Python and Scapy**

Extracted **domain names and IP addresses** from DNS traffic

Used **DNS cache and SNI** to identify domains in **encrypted HTTPS/QUIC traffic**

Displayed all results **in real time through a GUI**

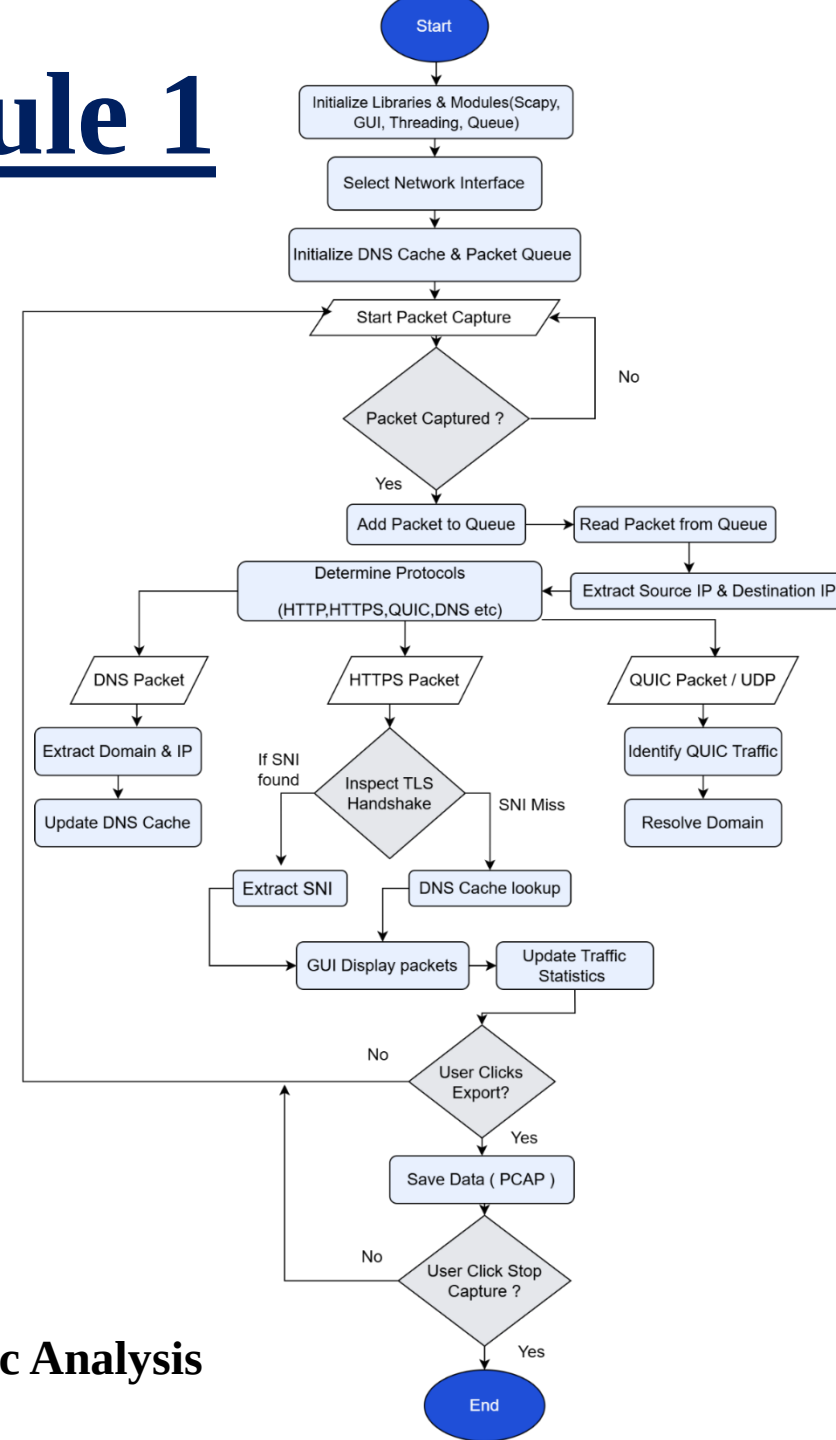
Design Methodology Module 1 [5/17]



DNS-Based Network Traffic Analysis

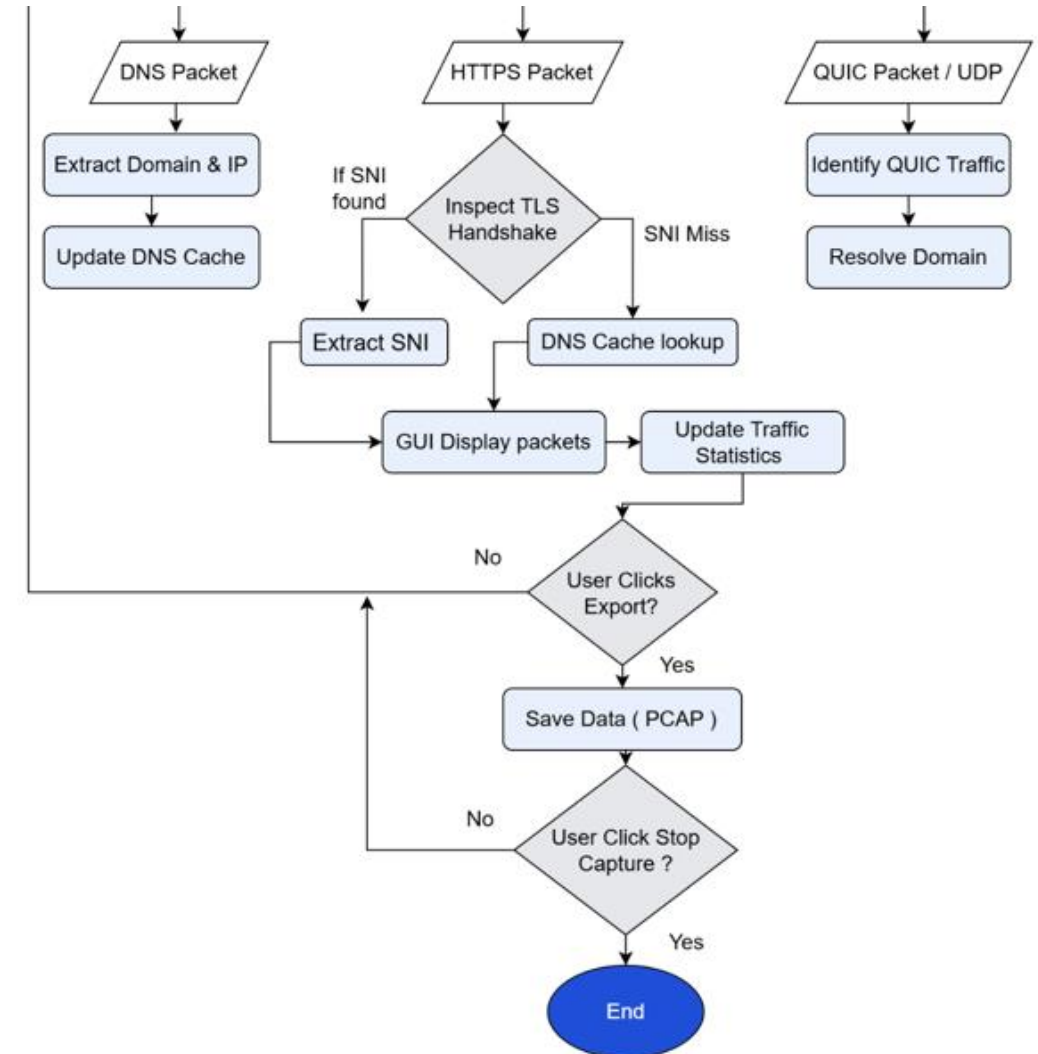
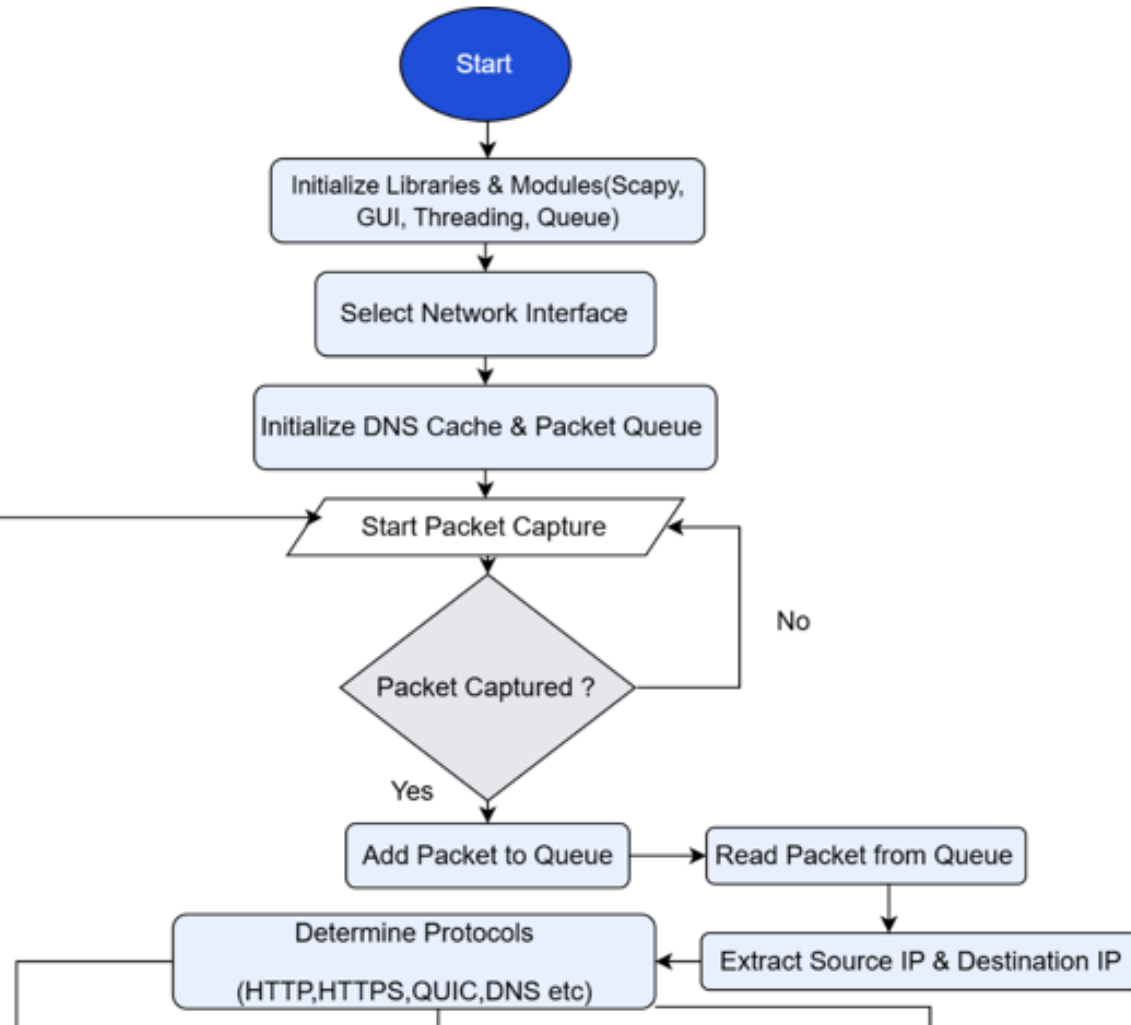
Design Methodology Module 1

[6/17]



DNS-Based Network Traffic Analysis

Design Methodology Module 1 [7/17]



Design Methodology Module 1 [8/17]:



URL Monitoring using MITM Proxy

In the second step, we used an MITM proxy to extract full URLs after decryption, allowing us to identify exact pages and short-form content.

Implemented **URL monitoring using a MITM proxy**

Intercepted traffic **after TLS decryption**

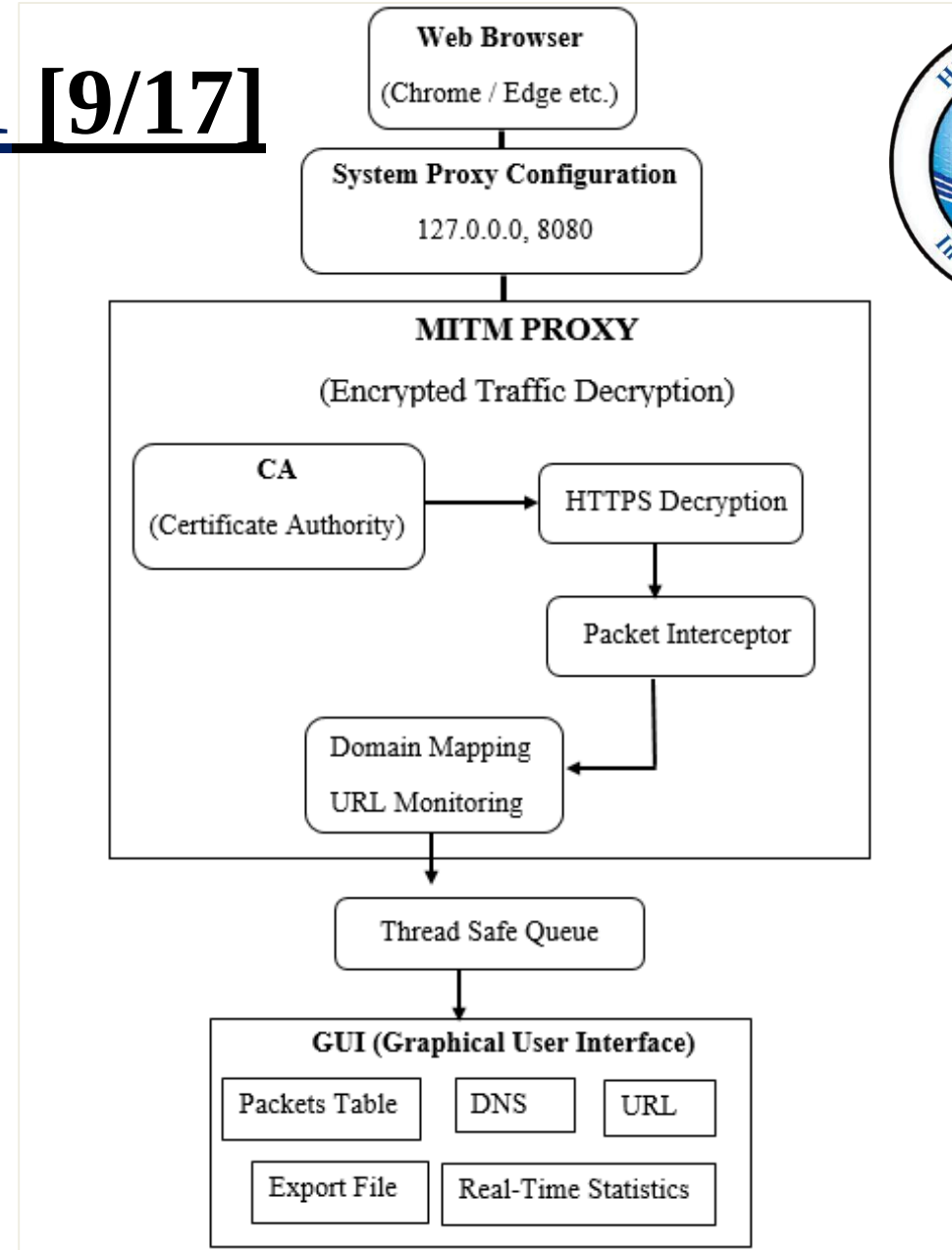
Extracted **full URLs and page-level information**

Identified **short-form video content** (e.g., YouTube Shorts, Instagram Reels)

Displayed URLs and related data **in real time through the GUI**

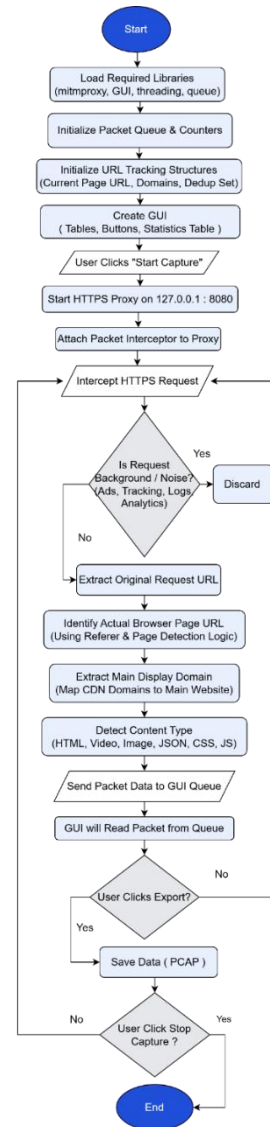


Design Methodology **Module 1** [9/17]

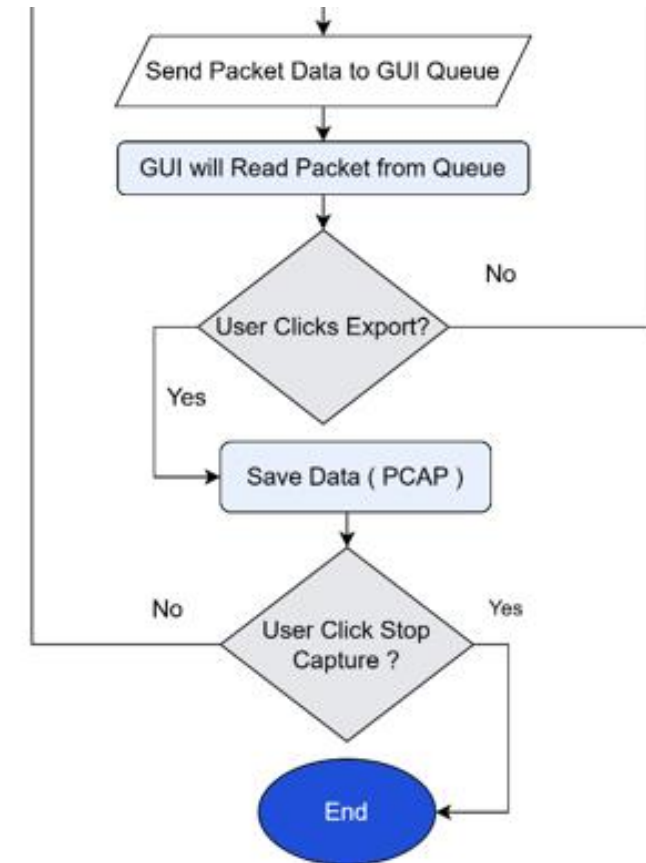
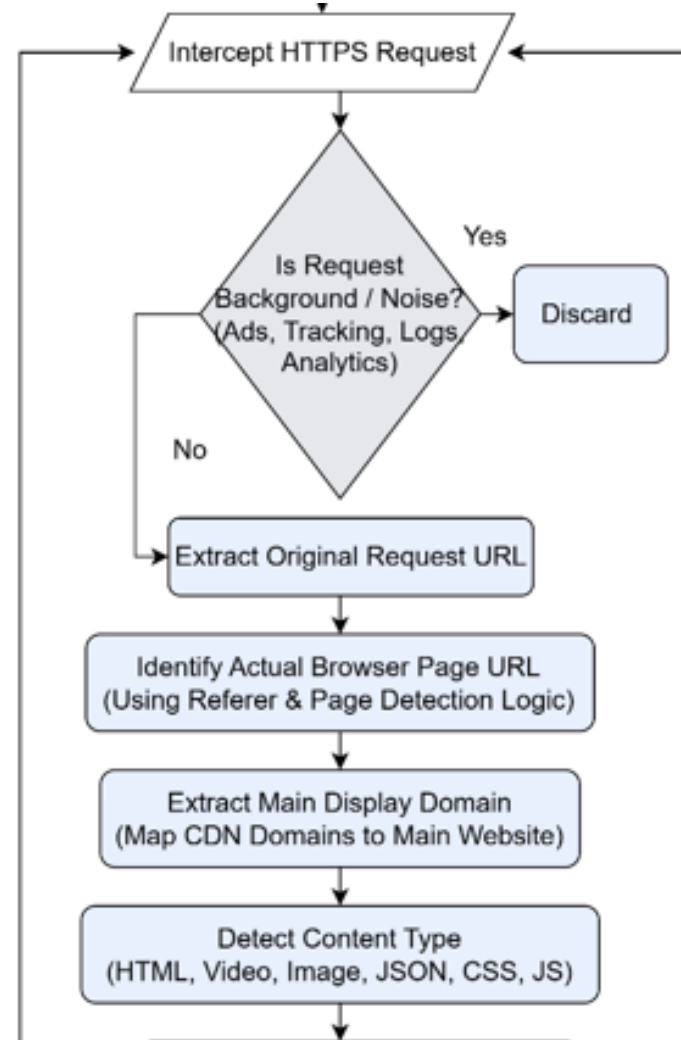
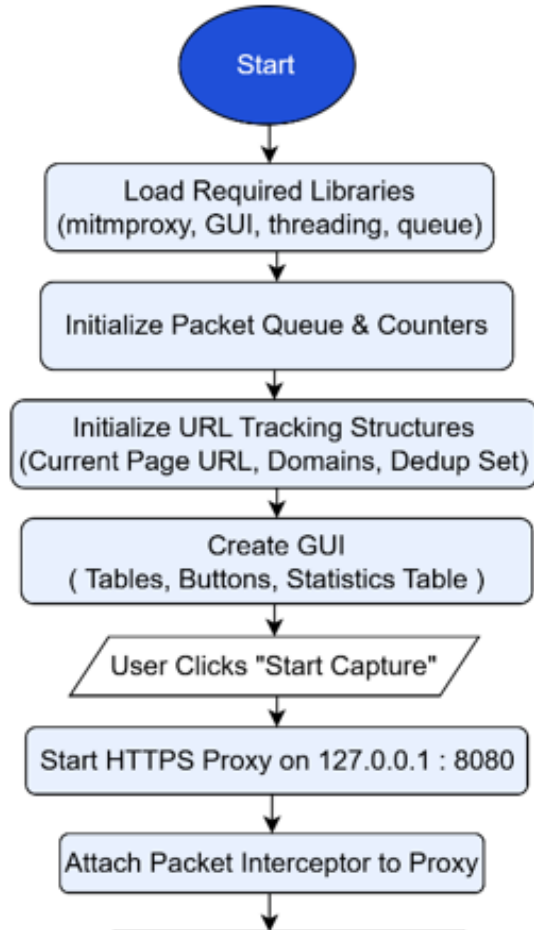


URL Monitoring Using MITM Proxy

Design Methodology Module 1 [10/17]



Design Methodology Module 1 [11/17]



Design Methodology Module 2 [12/17]



Raspberry Pi 5 was used as the core device to implement a network-level routing system.

Raspberry Pi OS was installed to support Linux-based networking features.

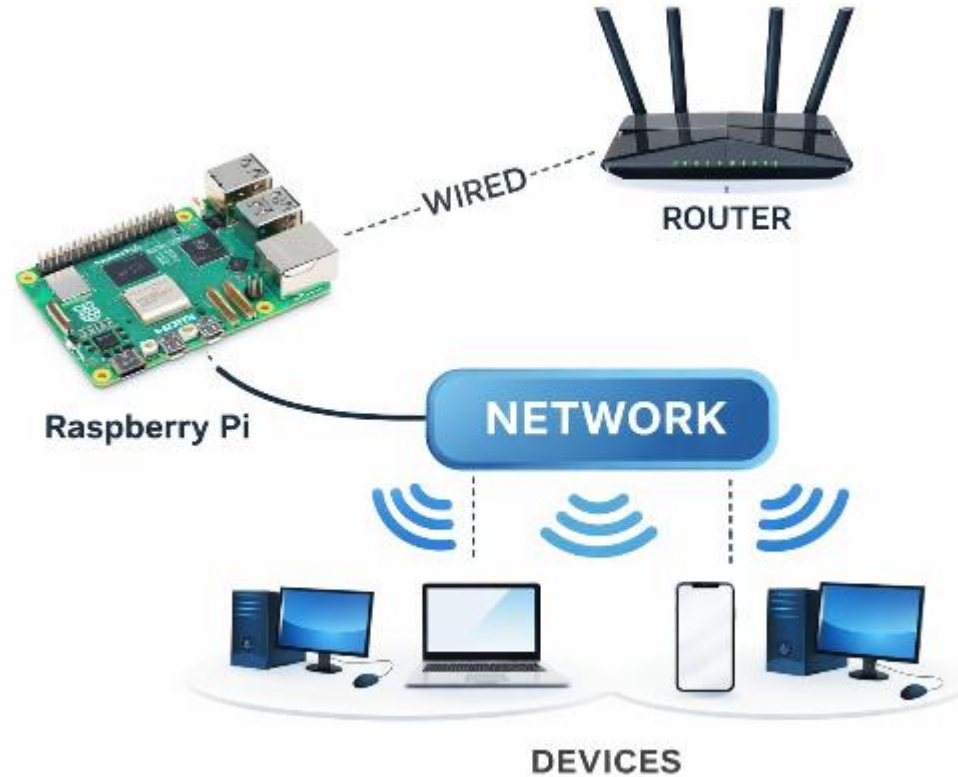
Remote configuration was enabled using VNC and SSH.

Internet was provided to the Raspberry Pi through Ethernet using laptop Internet Connection Sharing (ICS).

A secure Wi-Fi hotspot (PiRouter) was created on the Raspberry Pi.

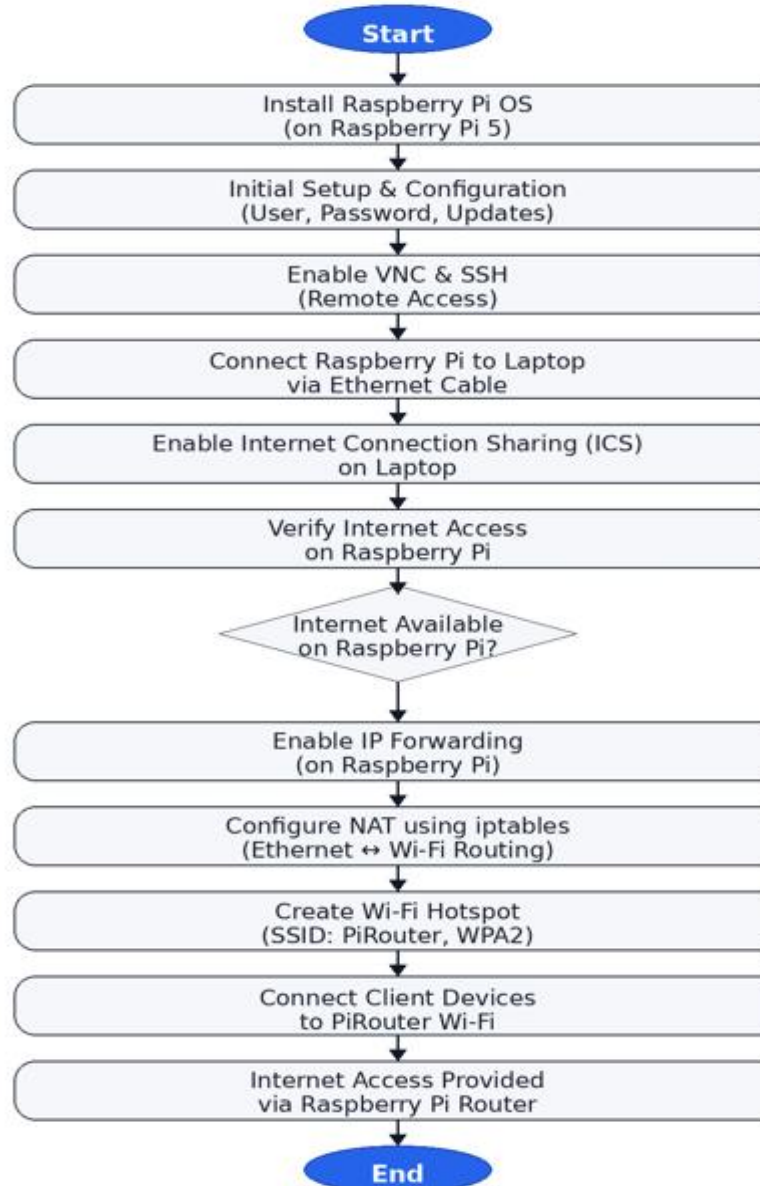
Client devices connected to the hotspot and accessed the internet via the Raspberry Pi router.

Design Methodology Module 2 [13/17]



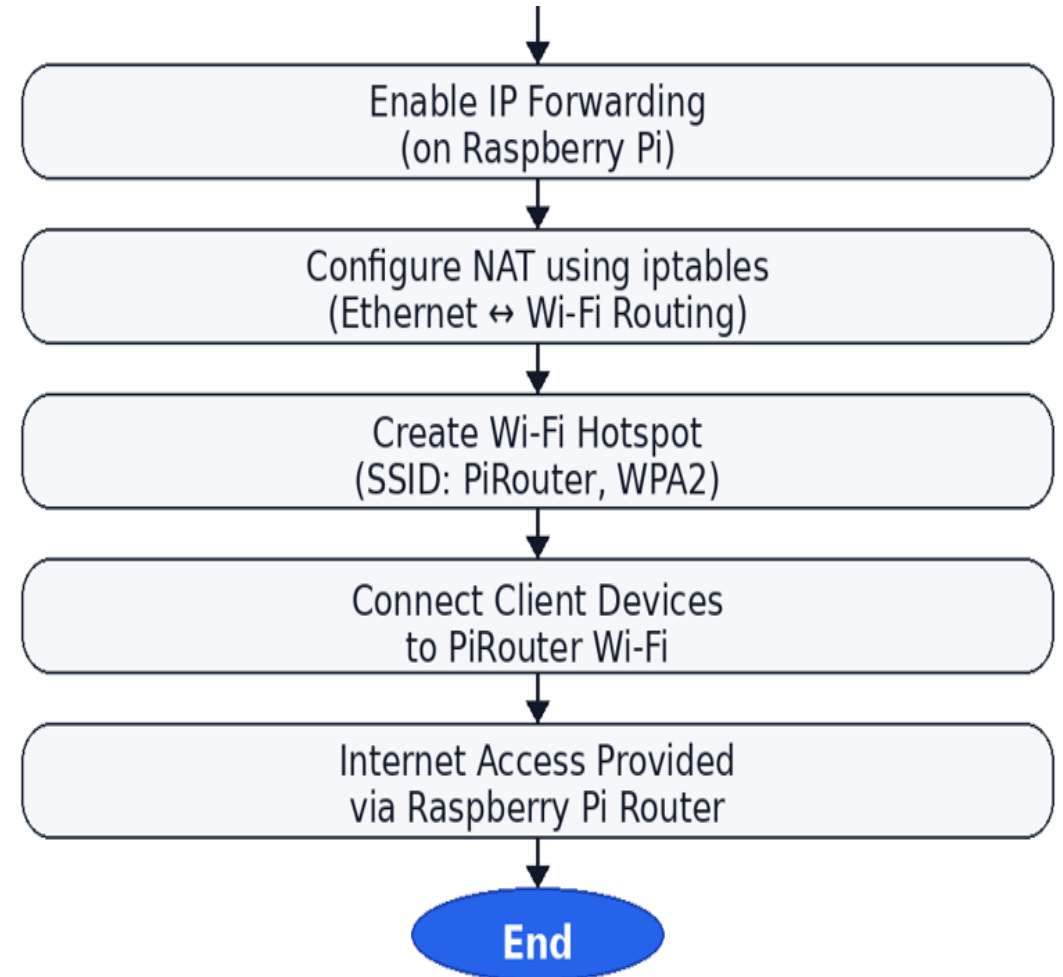
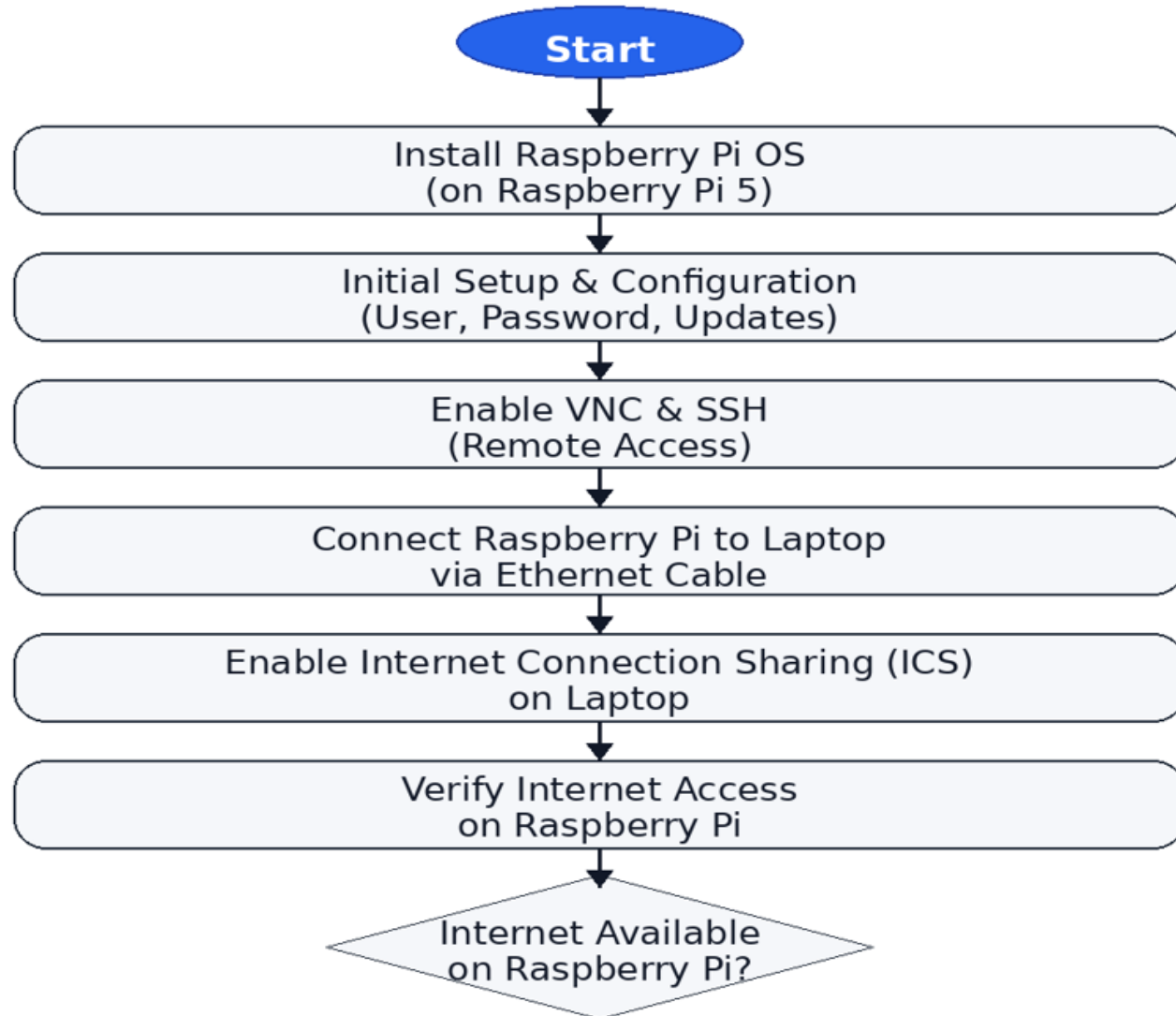
Raspberry pi5 Router Configuration

Design Methodology Module 2 [14/17]



Raspberry pi5 Router Configuration

Design Methodology Module 2 [15/17]





Implementation Module 1 [1/3]:

Programming & Development Tools

Python – Core programming language for system development

Scapy – Real-time packet capturing and network traffic analysis

mitmproxy – HTTPS Man-in-the-Middle proxy for URL monitoring

Tkinter – Graphical User Interface (GUI) development

Networking & System Tools

TLS / SNI HTTPS Protocols – For encrypted traffic handling

DNS Protocol – Domain name resolution and traffic analysis

QUIC (HTTP/3) – Encrypted UDP-based traffic identification

Multi-threading & Queue mechanisms – Smooth real-time processing

Implementation Module 2 [2/3]:



Raspberry pi5 Router Configuration

Install Raspberry Pi OS on Raspberry Pi 5

Perform initial setup and enable SSH

Connect Raspberry Pi to laptop via Ethernet

Enable Internet Connection Sharing (ICS) on laptop

Verify internet access on Raspberry Pi

Implementation Module 2 [3/3]:



Enable IP forwarding on Raspberry Pi

Configure IP on Real VNC Viewer

Create secure Wi-Fi hotspot (Pi Router)

Connect client devices to PiRouter

Internet successfully provided via Raspberry Pi router

Result Module 1 [1/13]:



DNS Monitoring System

Export Button

Start Button

Stop Button

Network Analyzer - Raspberry Pi 5

File

Interface: auto

Start

Stop

Clear

Stopped

Filters

Protocol: All

Direction: All

Reset

Statistics

IN: 3829

OUT: 2510

Total: 6339

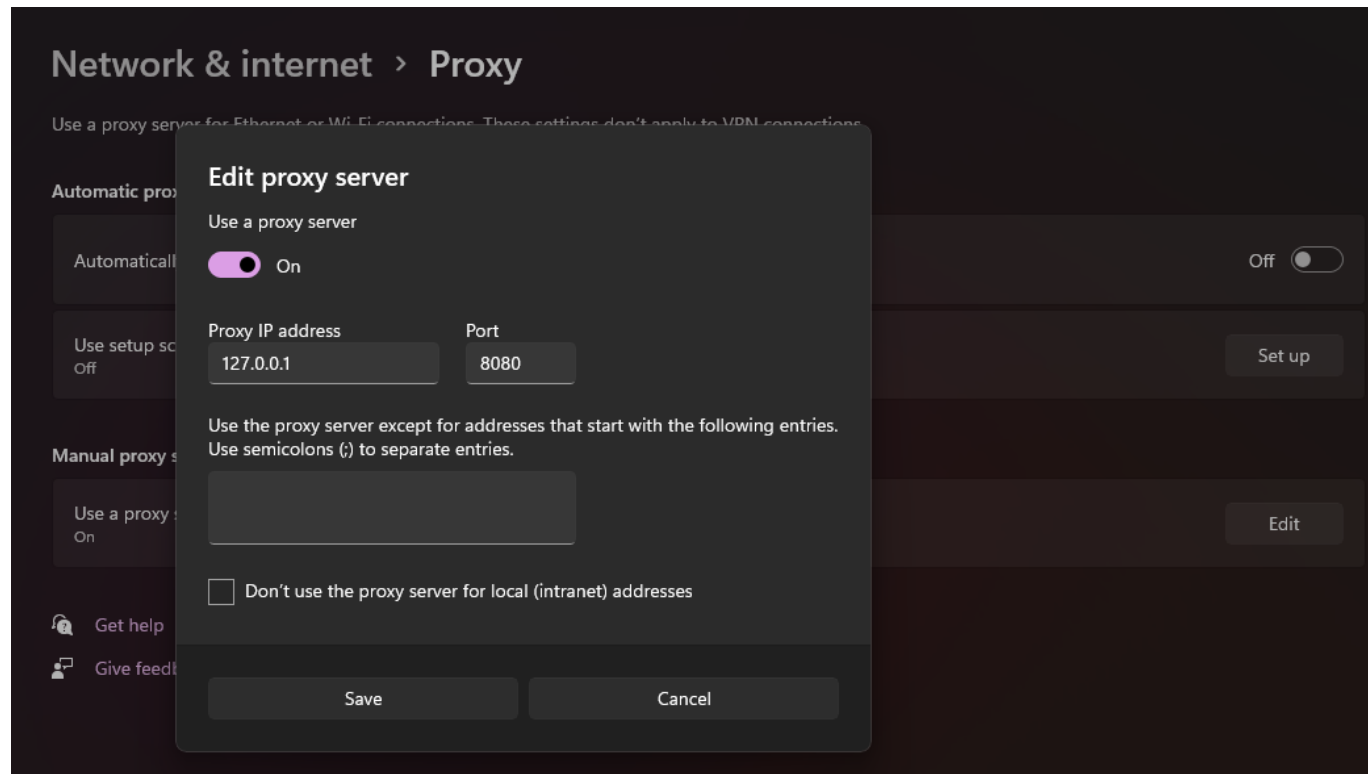
Packets

No	Time	Dir	Source	Dest	Protocol	Domain	Info
530	04:15:01.35	OUT	192.168.100.7	142.250.202.238	TCP		Port 54035→443
531	04:15:01.36	LOCAL	192.168.100.7	192.168.100.1	DNS	www.youtube.com	Query: www.youtube.com
532	04:15:01.36	LOCAL	192.168.100.7	192.168.100.1	DNS	www.youtube.com	Query: www.youtube.com
533	04:15:01.36	LOCAL	192.168.100.1	192.168.100.7	DNS	www.youtube.com	Response: www.youtube.com → b'youtube-ui.l.google.com.'
534	04:15:01.36	LOCAL	192.168.100.1	192.168.100.7	DNS	www.youtube.com	Response: www.youtube.com → b'youtube-ui.l.google.com.'
535	04:15:01.62	OUT	192.168.100.7	142.250.202.46	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
536	04:15:01.62	OUT	192.168.100.7	142.250.202.46	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
537	04:15:01.62	IN	142.250.202.238	192.168.100.7	TCP		Port 443→54035
538	04:15:01.62	OUT	192.168.100.7	142.250.202.238	TCP		Port 54035→443
539	04:15:01.62	OUT	192.168.100.7	142.250.202.238	HTTPS	youtube.com	youtube.com
540	04:15:01.62	OUT	192.168.100.7	142.250.202.238	HTTPS	youtube.com	youtube.com
541	04:15:01.63	IN	142.250.202.46	192.168.100.7	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
542	04:15:01.63	IN	142.250.202.46	192.168.100.7	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
543	04:15:01.63	IN	142.250.202.46	192.168.100.7	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
544	04:15:01.63	IN	142.250.202.46	192.168.100.7	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
545	04:15:01.63	OUT	192.168.100.7	142.250.202.46	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
546	04:15:01.63	IN	142.250.202.238	192.168.100.7	TCP		Port 443→54035
547	04:15:01.63	OUT	192.168.100.7	142.250.202.46	TCP		Port 54036→443
548	04:15:01.63	OUT	192.168.100.7	142.250.202.46	QUIC	www.youtube.com	HTTP/3 → www.youtube.com
549	04:15:01.63	IN	142.250.202.46	192.168.100.7	TCP		Port 443→54036

Result Module 1 [2/13]:



URL Monitoring using MITM Proxy



Proxy Will be Enabled.

Result Module 1 [3/13]:



When you start capturing by pressing start Button then we will be able to get domain name as well as URL of the packet.

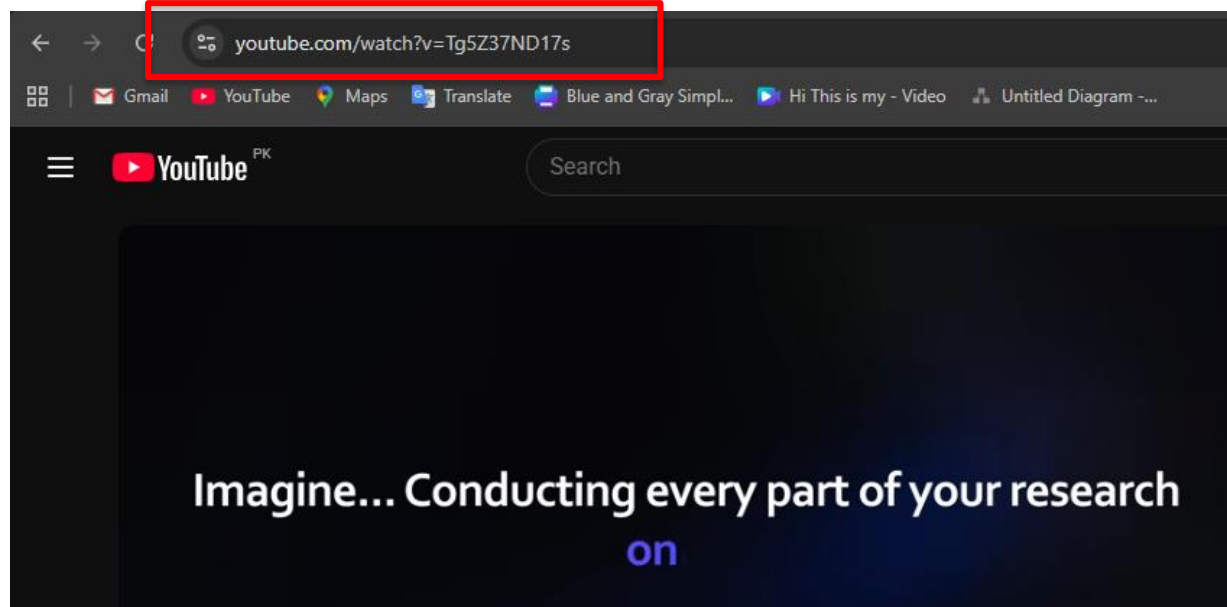
Professional HTTPS Packet Interceptor					
▶ Start Capture Stop Capture Export Data Clear All Setup Guide Certificate Folder Stopped					
Packets: 456 Top Domains: www.youtube.com, i.ytimg.com, static.canva.com					
Captured Packets					
No	Time	Method	Type	Domain	URL
1	04:02:50.772	GET	HTTP Request	clientservices.googleapis.com	https://clientservices.googleapis.com/chrome-variations/seed?osname=win&channel=stable&milestone=143
2	04:02:50.793	GET	HTTP Request	www.microsoft.com	http://www.microsoft.com/mitmproxy-352809477949792998396289060453647071536262098974.crl
3	04:02:51.332	POST	HTTP Request	checkappexec.microsoft.com	https://checkappexec.microsoft.com/windows/shell/actions
4	04:02:51.402	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=&oit=0&oft=...
5	04:02:51.405	GET	HTTP Request	www.google.com	https://www.google.com/async/newtab_ogb?hl=en-US&async=fixed:0
6	04:02:51.407	GET	HTTP Request	www.google.com	https://www.google.com/async/ddljson?async=ntp:2
7	04:02:51.472	GET	HTTP Request	www.google.com	https://www.google.com/async/folae?async=_fmt:pb
8	04:02:51.730	GET	HTTP Request	www.google.com	https://www.google.com/async/newtab_promos
9	04:02:51.760	POST	HTTP Request	www.googleapis.com	https://www.googleapis.com/oauth2/v4/token
10	04:02:51.875	POST	HTTP Request	accounts.google.com	https://accounts.google.com/ListAccounts?gpsia=1&source=ChromiumBrowser&json=standard&laf=b64bin
11	04:02:52.190	GET	HTTP Request	lh3.google.com	https://lh3.google.com/u/0/ogw/AF2bZyi6x9IWBi_IWPTJVFx6biddioNGsazF3jOOQnPCeh3HT-g=s32-c-mo
12	04:02:52.230	GET	HTTP Request	kidsmanagement-pa.googleapis.com	https://kidsmanagement-pa.googleapis.com/kidsmanagement/v1/families/mine/members?alt=proto&allow_emp...
13	04:02:52.288	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=&oit=0&gs_r...
14	04:02:52.289	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=&oit=0&oft=...
15	04:02:52.401	OPTIONS	CORS Preflight	ogads-pa.clients6.google.com	https://ogads-pa.clients6.google.com/\$rpc/google.internal.onegoogle.asyncdata.v1.AsyncDataService/Ge...
16	04:02:52.636	GET	HTTP Request	lh3.googleusercontent.com	https://lh3.googleusercontent.com/rd-ogw/AF2bZyjk23-vFPXveUoXzQse32UULsefyXUW-1n7BFYGzFY9dU8RYV6WRp...
17	04:02:52.907	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1&cp=...
18	04:02:53.036	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1&cp...
19	04:02:53.310	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1&c...
20	04:02:53.497	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1&...
21	04:02:53.599	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1...
22	04:02:53.825	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1...
23	04:02:54.022	GET	HTTP Request	www.google.com	https://www.google.com/complete/search?client=chrome-omni&gs_ri=chrome-ext-ansg&xssi=t&q=y&oit=1...
24	04:02:54.086	POST	HTTP Request	cloudsearch.googleapis.com	https://cloudsearch.googleapis.com/v1/query/search
25	04:02:54.463	GET	HTTP Request	encrypted-tbn0.gstatic.com	https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQKIE7anyEoBkUcZRbeSGkBvaFLd9IXmqTC-mgDpsc&is=10

Result Module 1 [4/13]:



Here we are able to get the exact page URL that is shown in Browser in real time.

Domain	URL
www.youtube.com	https://rr4---sn-jtcxg-gf2s.googlevideo.com/generate_204
www.youtube.com	https://www.youtube.com/watch?v=Tg5Z37ND17s
www.youtube.com	https://www.youtube.com/watch?v=Tg5Z37ND17s
www.youtube.com	https://rr4---sn-jtcxg-gf2s.googlevideo.com/videoplayback?expire=1767722584&ei=-PlcaaSIBqiq-_UPiqf6m...
www.youtube.com	https://rr4---sn-jtcxg-gf2s.googlevideo.com/videoplayback?expire=1767722603&ei=C_pcaZbADua_i9oPwpGC6...
www.youtube.com	https://rr3---sn-jtcxg-gf2s.googlevideo.com/videoplayback?expire=1767722603&ei=C_pcadmPlua_i9oPwpGC6...

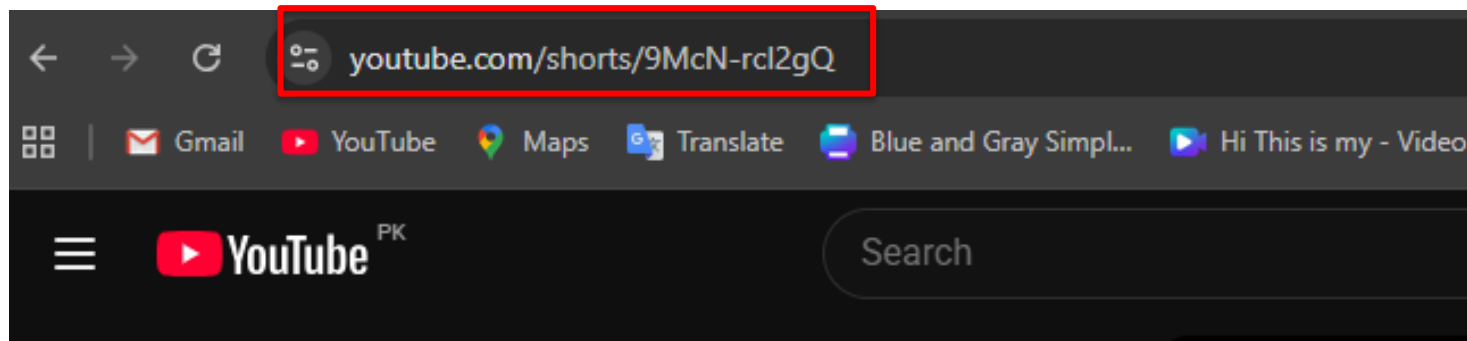


Result Module 1 [5/13]:



Here we are able to get the exact page URL that is shown in Browser in real time.

Domain	URL
www.youtube.com	https://rr3---sn-jtcxg-gf2s.googlevideo.com/videoplayback?expire=1767722680&ei=WPpcaf38DKK9zPsPhs6wm...
www.youtube.com	https://www.youtube.com/shorts/QMRh3oE5BaU
www.youtube.com	https://www.youtube.com/shorts/9McN-rcI2gQ
i.ytimg.com	https://i.ytimg.com/vi/9McN-rcI2gQ/hqdefault.jpg?sqp=-oaymwFBCNACELwBSFryq4qpAzMIARUAAIhCGADYAQHiAQo...
yt3.ggpht.com	https://yt3.ggpht.com/YwCUug784ZEVNG7Lu4d_qmsKNt0UPCjMKMQMfXbhXiJhVBGY4iSV1Kk68PONmmKkd6G0ht9JA=s48...
www.youtube.com	https://rr2---sn-jtcxg-gf2z.googlevideo.com/videoplayback?expire=1767722684&ei=XPpcaYTaK_CMkdUPwbra8...
www.google.com	https://www.google.com/async/ddljson?async=ntp:2
www.google.com	https://www.google.com/async/newtab_promos

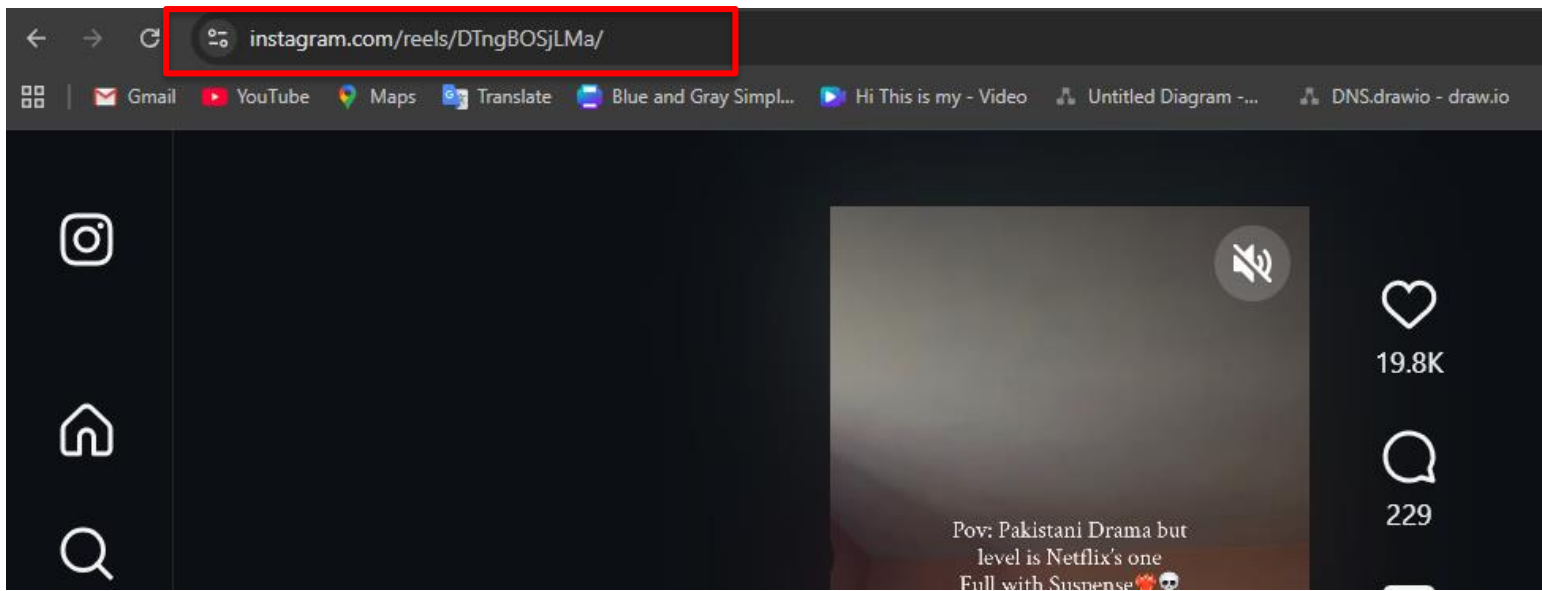




Result Module 1 [6/13]:

Here we are able to get the exact page URL that is shown in Browser in real time.

11	08:53:46.513	GET	HTTP Request	www.instagram.com	https://instagram.fisb34-1.fna.fbcdn.net/o1/v/t2/f2/m367/AQNE-RbYSwrUluuhD_V02roMaAtIG1cScYPDIc6HCh...
12	08:53:49.350	GET	HTTP Request	edge-chat.facebook.com	https://edge-chat.facebook.com/chat?sid=7903017900703743&cid=6f71af0d-06be-4fa5-bd4c-2e703f2cce88
13	08:53:49.831	POST	HTTP Request	edge-chat.facebook.com	https://edge-chat.facebook.com/mqtt/pull?sid=4893092616339455&chunked=true
14	08:53:59.685	POST	HTTP Request	www.instagram.com	https://www.instagram.com/reels/DTngBOSjLMa/
15	08:54:00.274	GET	HTTP Request	www.instagram.com	https://www.instagram.com/reels/DTngBOSjLMa/
16	08:54:00.973	POST	HTTP Request	www.instagram.com	https://www.instagram.com/reels/DTngBOSjLMa/
17	08:54:01.412	GET	HTTP Request	media-mct1-2.cdn.whatsapp.net	https://media-mct1-2.cdn.whatsapp.net/v/t62.7117-24/563117330_1509991083420701_5379769731596904927_n...



Result Module 2 [7/13]:



Here Raspberry pi5 OS Customization

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

pi

Password:

Saved (hidden) — leave blank to keep

pi123

Confirm password:

Re-enter to change password

pi123

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

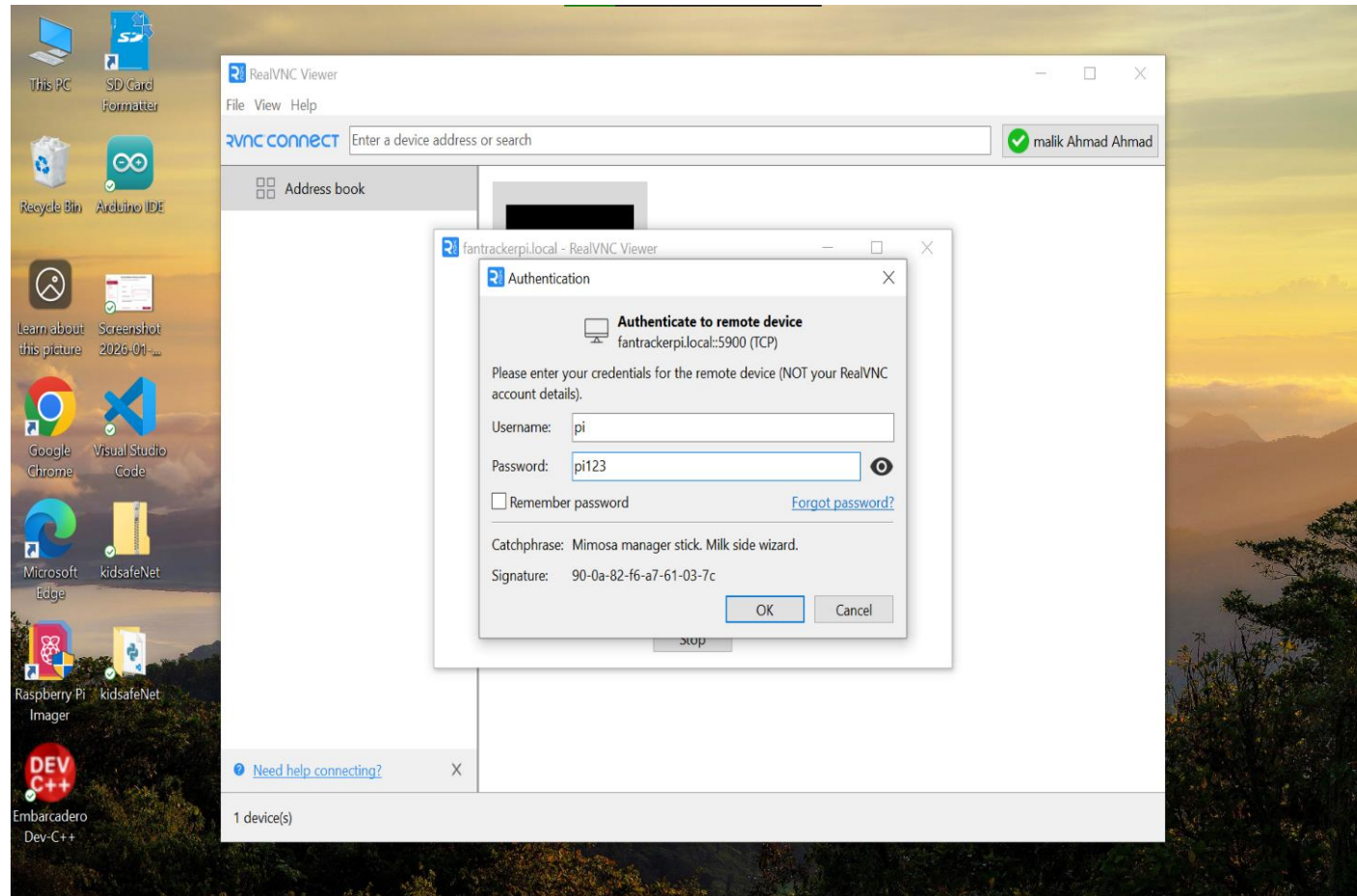
BACK

NEXT

Result Module 2 [8/13]:



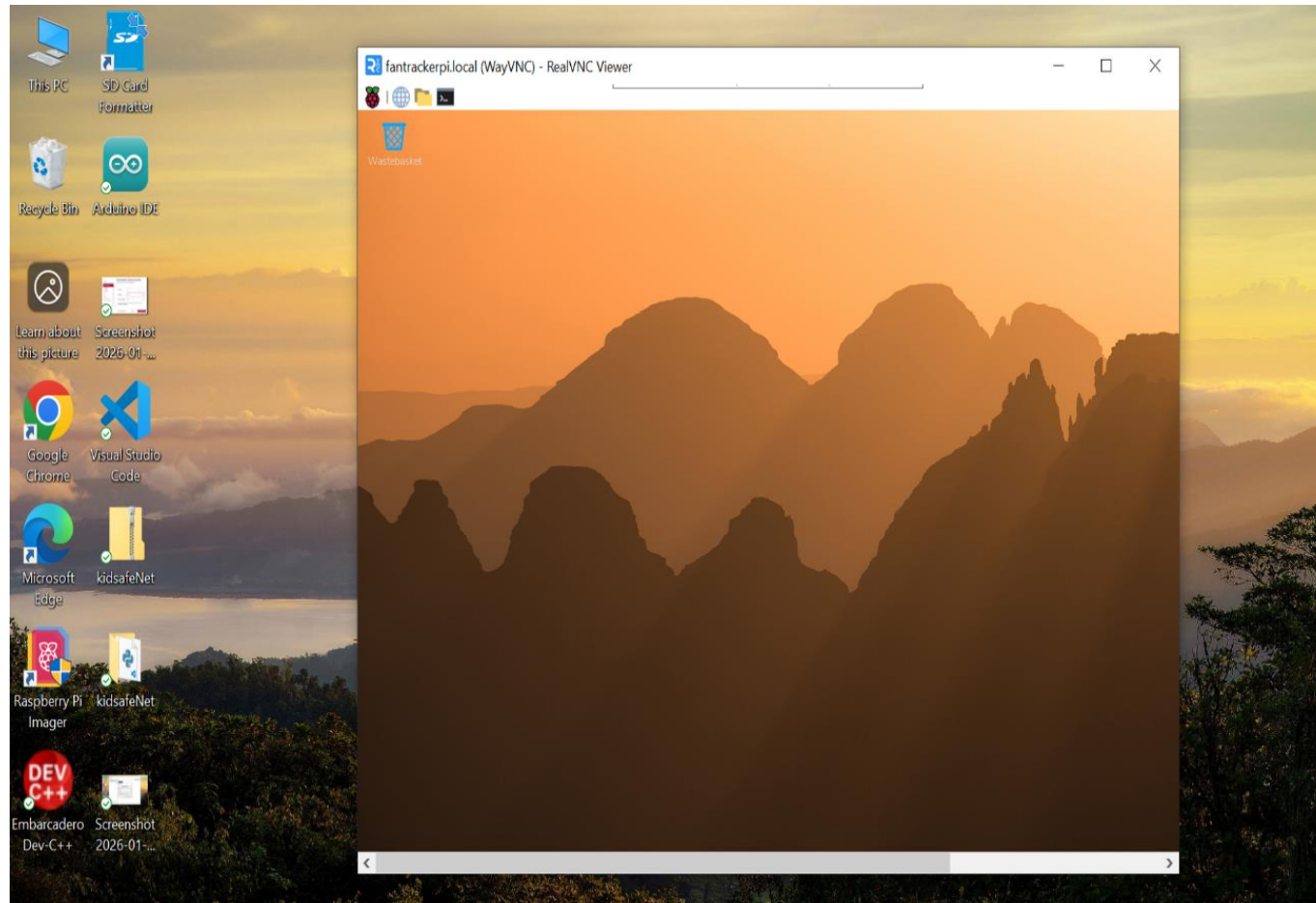
Here Pi5 is connected with Real VNC Viewer



Result Module 2 [9/13]:



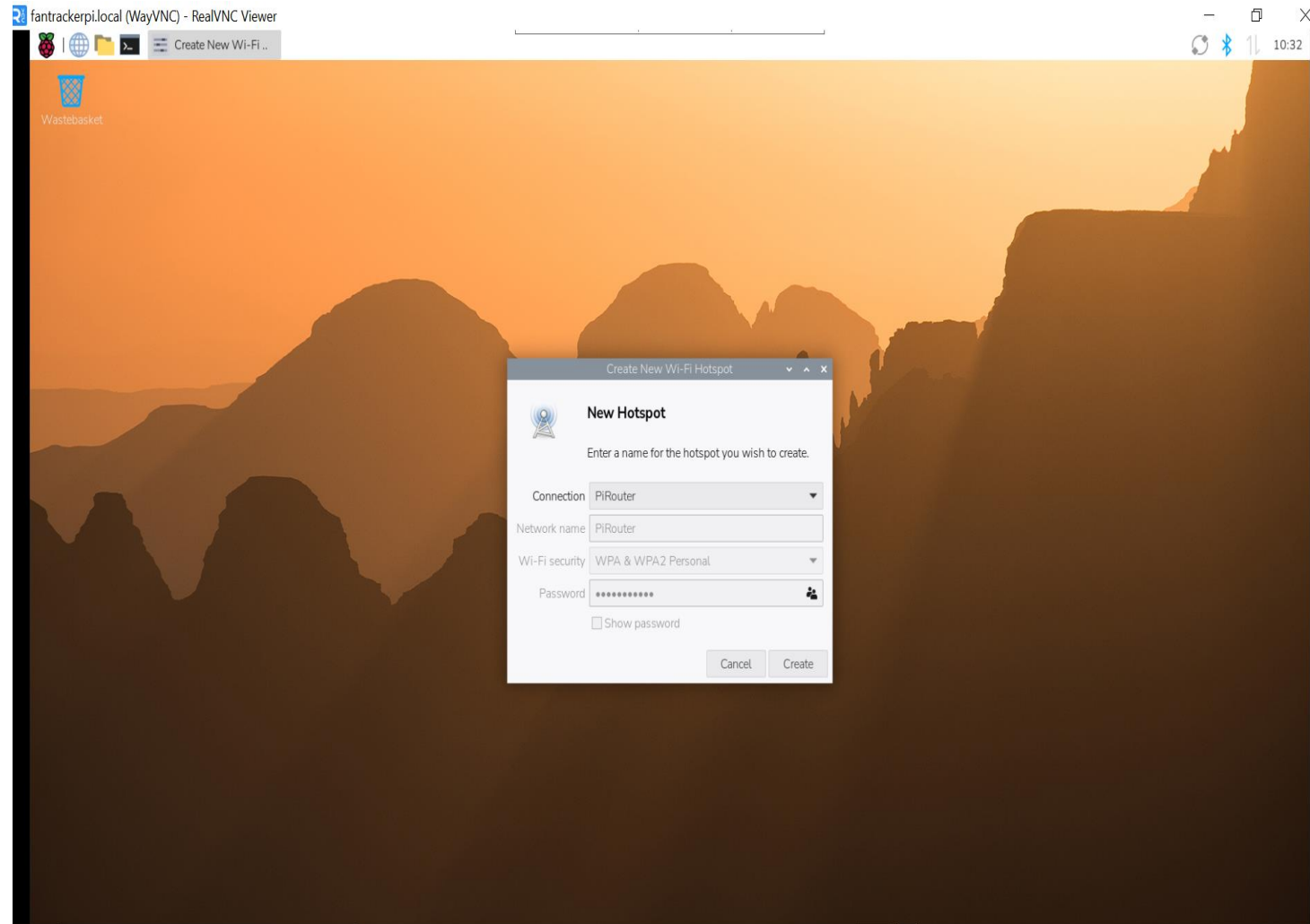
Here Pi5 Interface on Real VNC Viewer



Result Module 2 [10/13]:



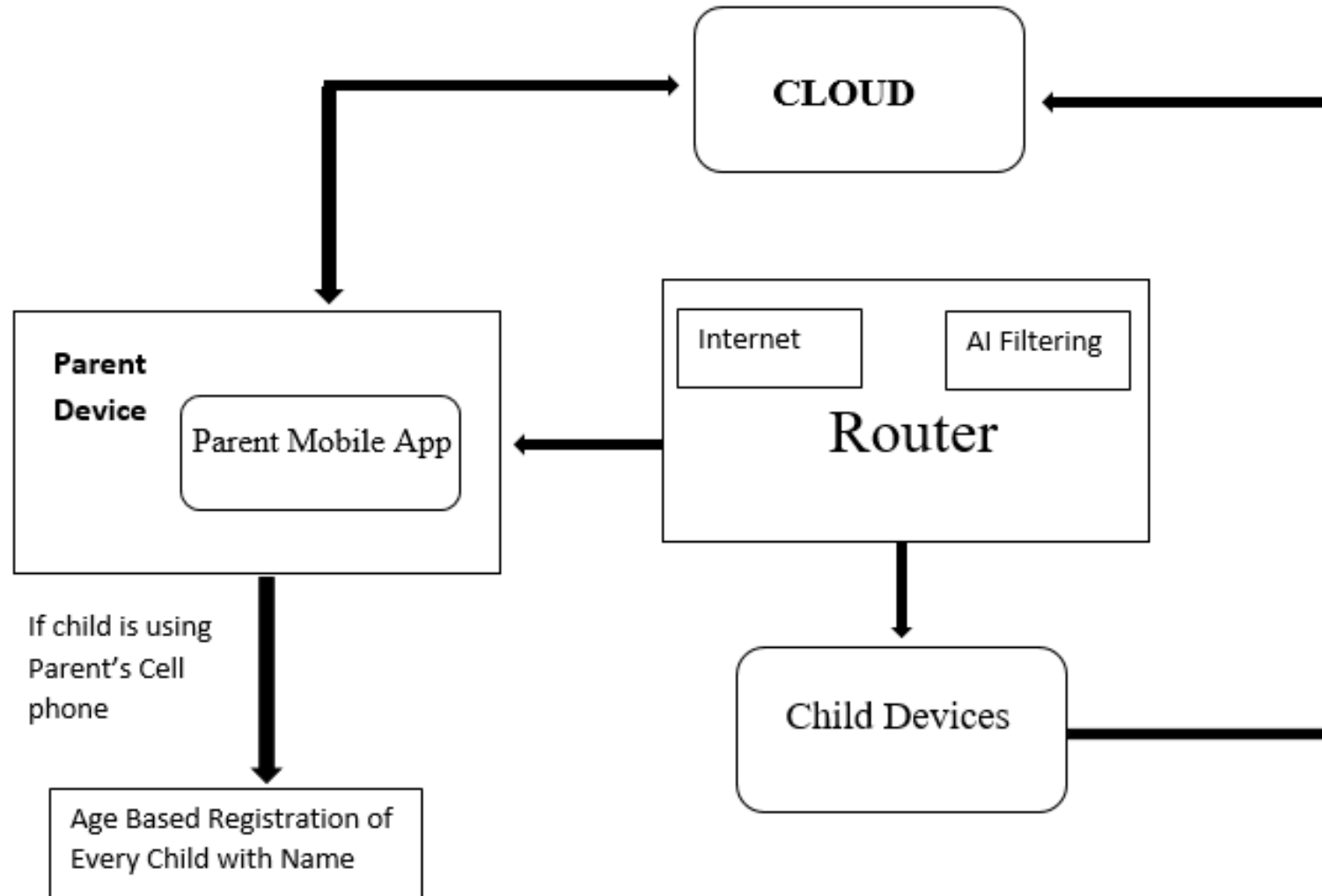
Here Pi5 Configure as a Router





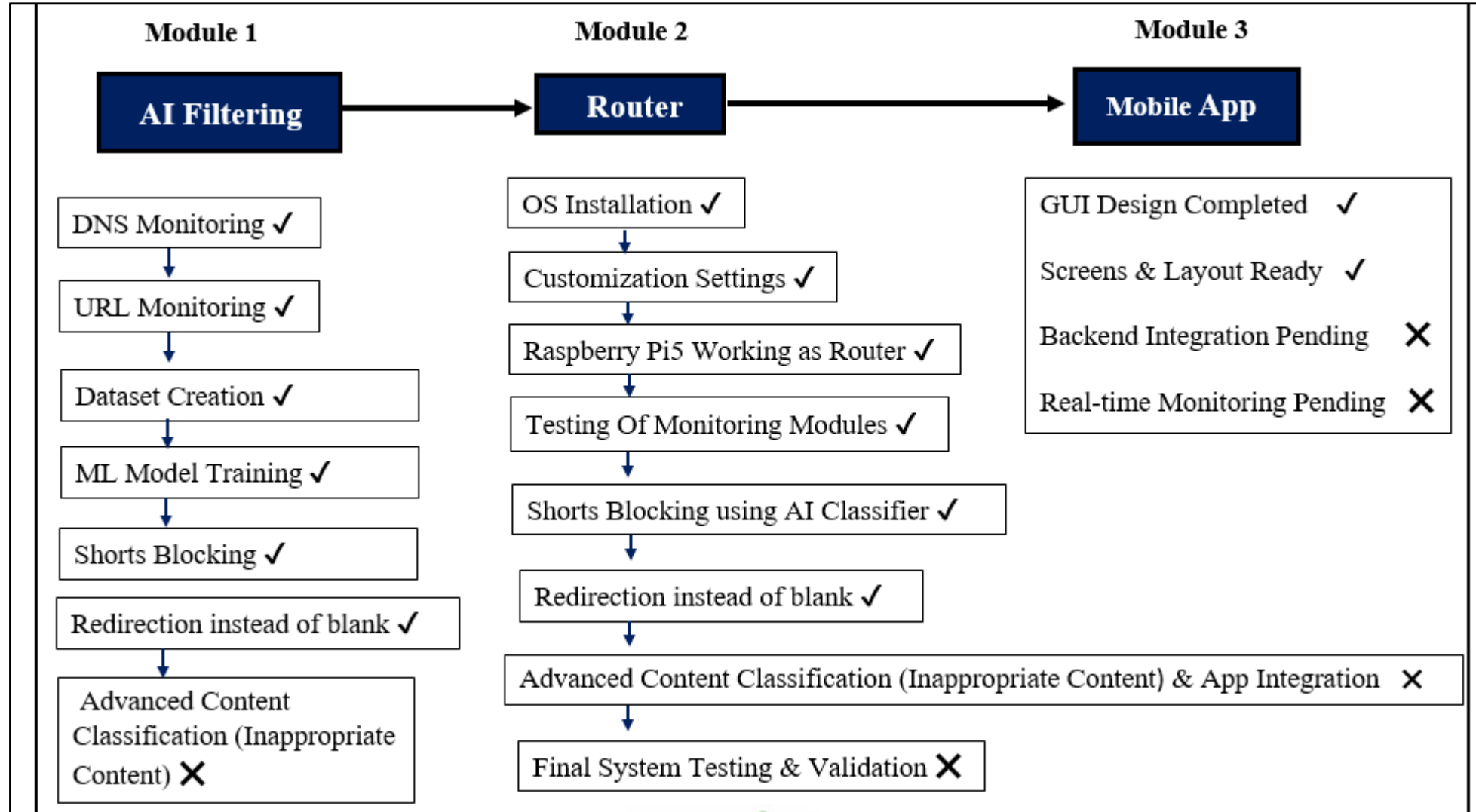
Progress Presentation 2

Design Methodology [1/9]



AI Based Parental Control Router

Design Methodology [2/9]



Design Methodology [3/9]



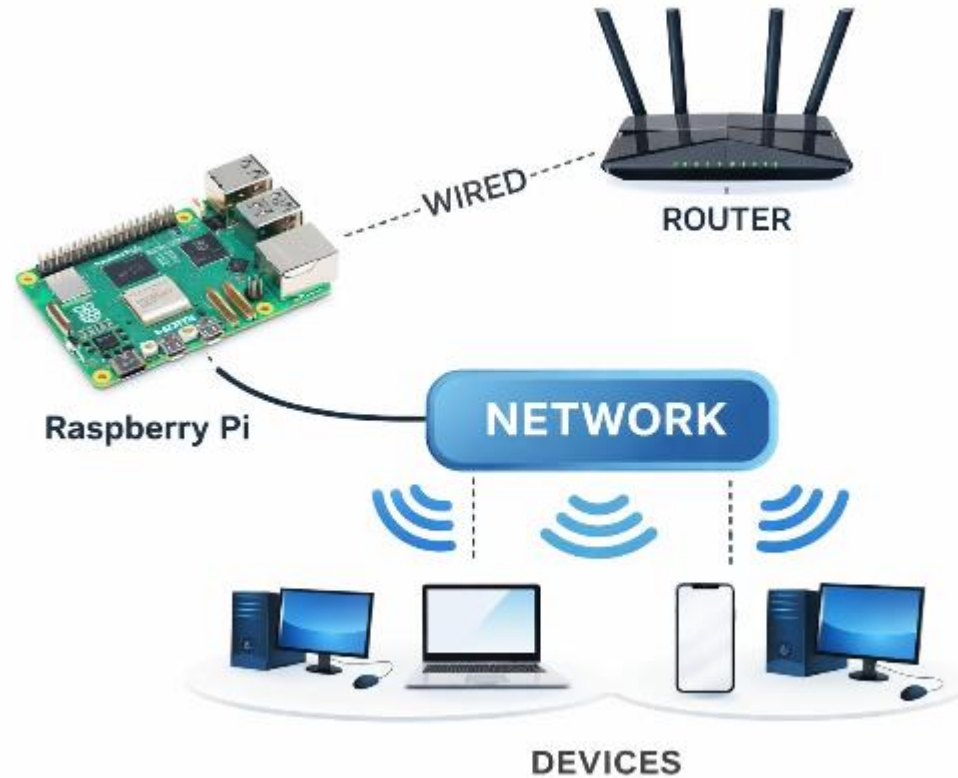
Previous Phase [DNS and URL Monitoring on Local System]

DNS-based traffic monitoring implemented

HTTPS URL monitoring using MITM proxy

System was running on local machine (laptop)

Design Methodology [4/9]



Raspberry pi5 Router Configuration

Design Methodology [5/9]



Migration of **URL** Monitoring Module to Raspberry Pi 5

Modules shifted from laptop to Raspberry Pi 5

Raspberry Pi configured as router

Devices connected through Pi Wi-Fi

All traffic passes through Pi

System moved from monitoring tool → real network device

Design Methodology [6/9]



Dataset Generation & AI Classifier Training

Dataset generated from captured traffic

ML models trained using Google Colab

Multiple classifiers tested

CatBoost selected as best model

Accuracy: 89.35%

AUC: 93.06%

Classifier	Accuracy	CV Score	AUC
1 CatBoost	89.35%	91.61%±0.87	93.06%
2 Random Forest	88.59%	90.66%±1.36	92.89%
3 Gradient Boosting	88.59%	90.66%±1.18	92.01%
Logistic Regression	88.21%	87.80%±1.48	89.48%
SVM (RBF)	88.21%	90.37%±1.54	92.73%
XGBoost	88.21%	90.75%±1.06	93.26%
Decision Tree	87.83%	89.14%±2.96	87.93%
SVM (Linear)	86.69%	88.66%±1.59	91.19%

System can now understand traffic behavior.

Design Methodology [7/9]



Short Form Videos Blocking & Redirecting Mechanism

AI model classifies traffic in real time

If Shorts Detects it will be Blocked

Instead of blocking with black screen

System redirects to safe content

Control is now implemented at network level

Design Methodology [8/9]



Mobile APP

Parent-managed, safety-focused system with secure authentication and role-based access.

Age-based content filtering, screen-time tracking, safety analysis, and activity logging.

Parental controls with approvals, alerts, and bilingual support (English & Urdu).

Built using Flutter (Dart SDK), and Java/JavaScript integration.

Design Methodology [9/9]



KidsafeNet is a child-safe digital platform. Connects parents and children using secure email-based accounts.

Provides age-appropriate and safe content.

Supports dual-language learning

Ensures a protected digital environment for children

Implementation [1/6]:

Raspberry pi5 Router Configuration



Install Raspberry Pi OS on Raspberry Pi 5

Perform initial setup and enable SSH

Connect Raspberry Pi to laptop via Ethernet

Enable Internet Connection Sharing (ICS) on laptop

Verify internet access on Raspberry Pi

Implementation [2/6]:



Enable IP forwarding on Raspberry Pi

Configure IP on Real VNC Viewer

Create secure Wi-Fi hotspot (Pi Router)

Connect client devices to PiRouter

Internet successfully provided via Raspberry Pi router



Implementation [3/6]:

Migration of Monitoring Modules to Raspberry Pi

URL monitoring (MITM proxy) configured on Pi

Required libraries and dependencies installed

- Configured MITM proxy on port 8080
- Installed certificate on client devices
- Verified HTTPS interception

Monitoring system successfully moved from local machine to network device.



Implementation [4/6]:

Dataset Generation & Model Training

Traffic captured using monitoring modules

Dataset created from real browsing activity

Features extracted:

- URL patterns
- Request headers
- Response size
- Content type

ML Training:

- Models trained on Google Colab
- CatBoost selected as best model



Implementation [5/6]:

CatBoost (Best Model)

A powerful boosting algorithm designed for structured data

How it works:

Builds trees sequentially like boosting

Learns from previous mistakes

Handles sparse and complex data efficiently

Reduces overfitting automatically



Implementation [6/6]:

AI-Based Shorts Blocking & Redirection Implementation

Model classifies traffic in real time

If prediction = Shorts / TikTok → block request

Replace URL with safe video/content

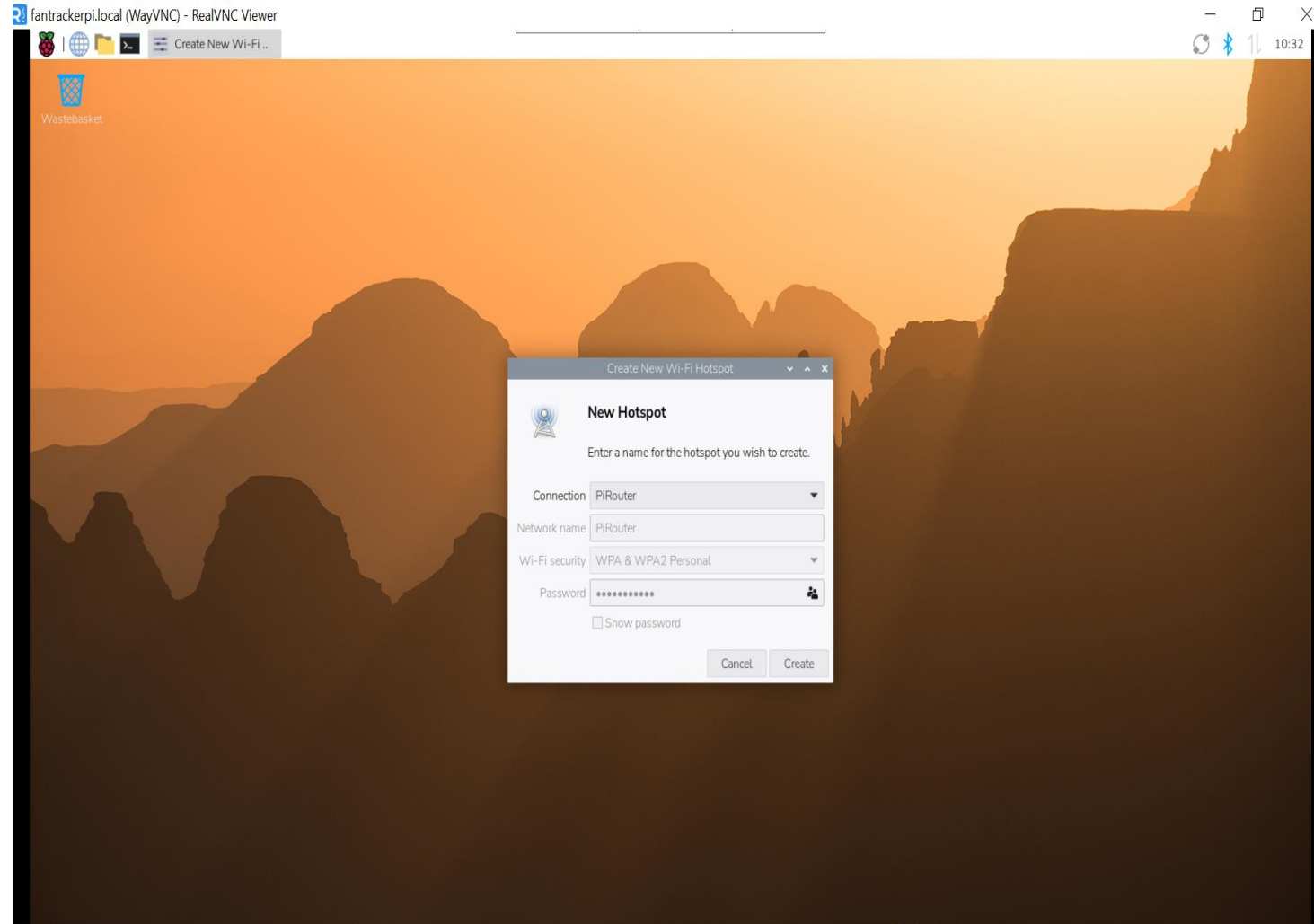
Send modified response

System performs automatic network-level blocking.

Result [1/10]:



Here Pi5 Configure as a Router



Result [2/10]:

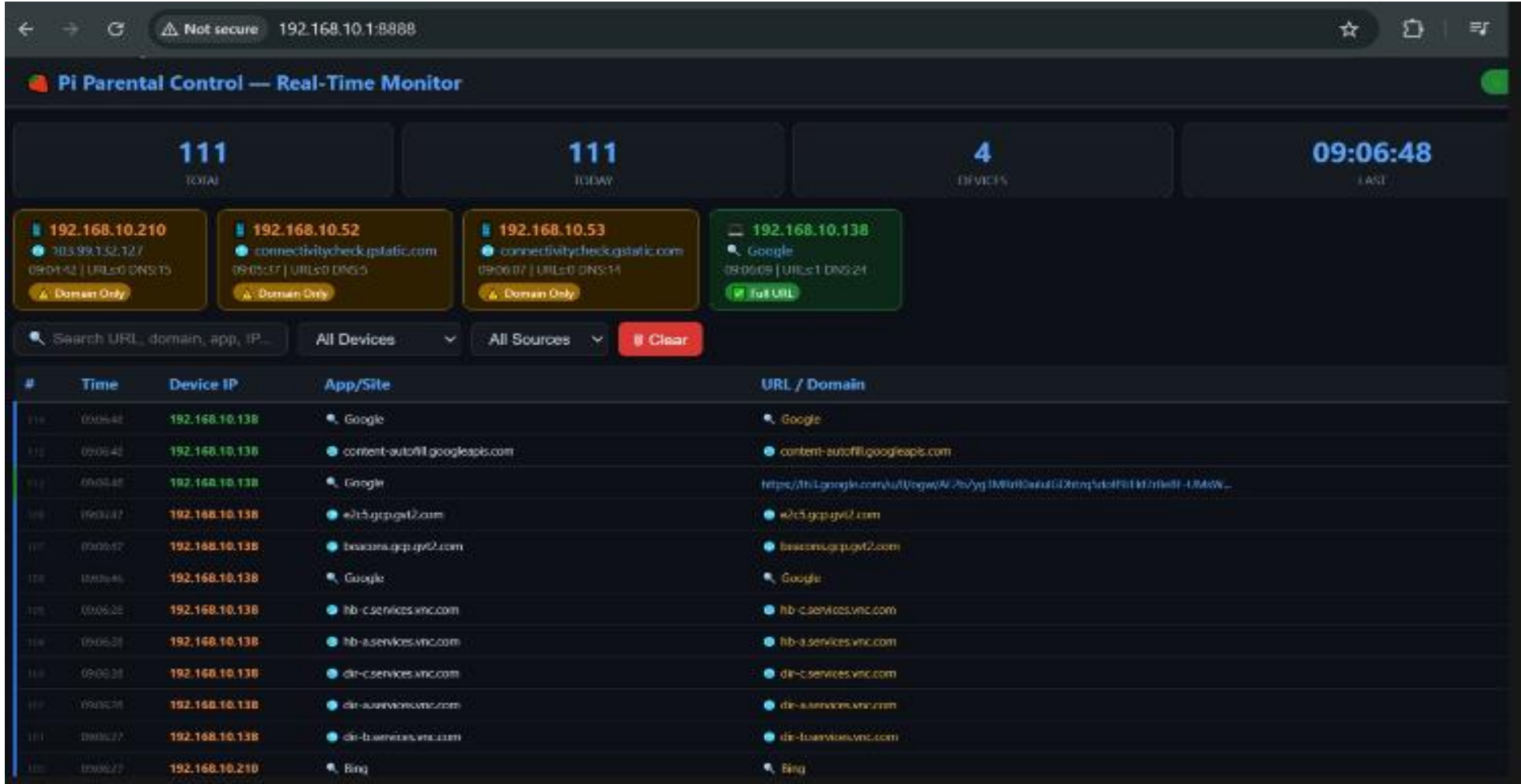
Here other devices connected with Pi5 Wi-Fi



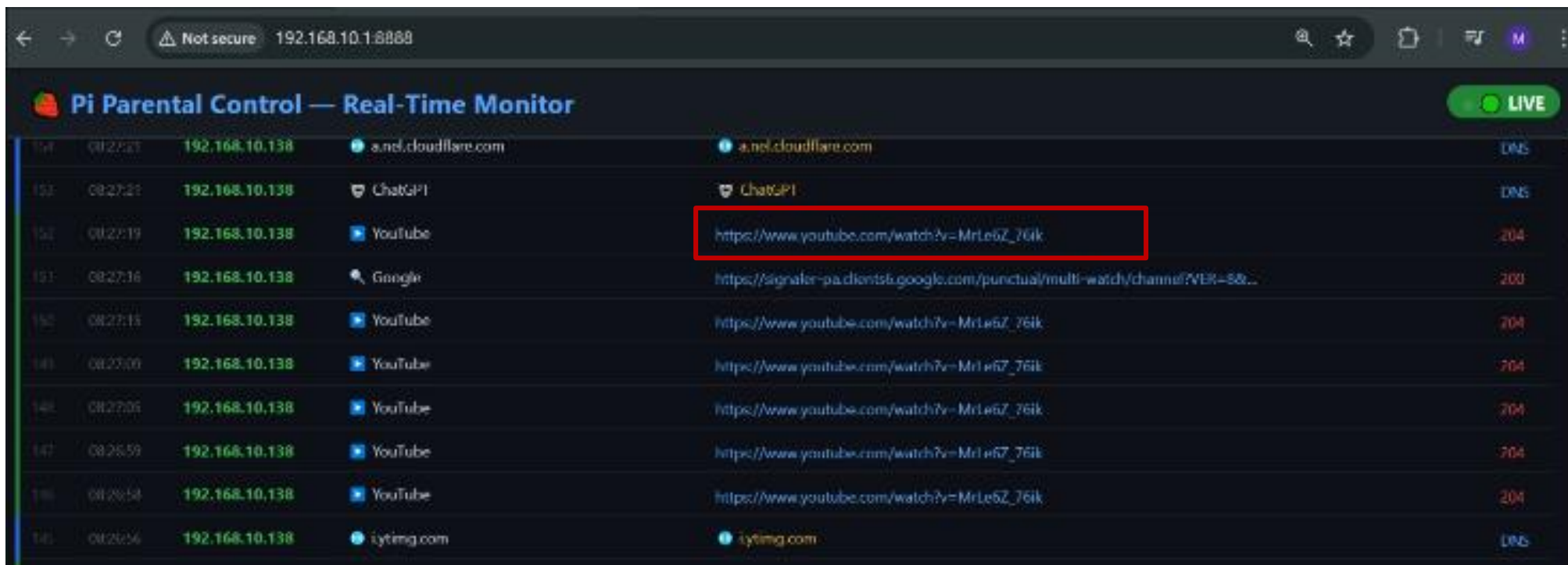
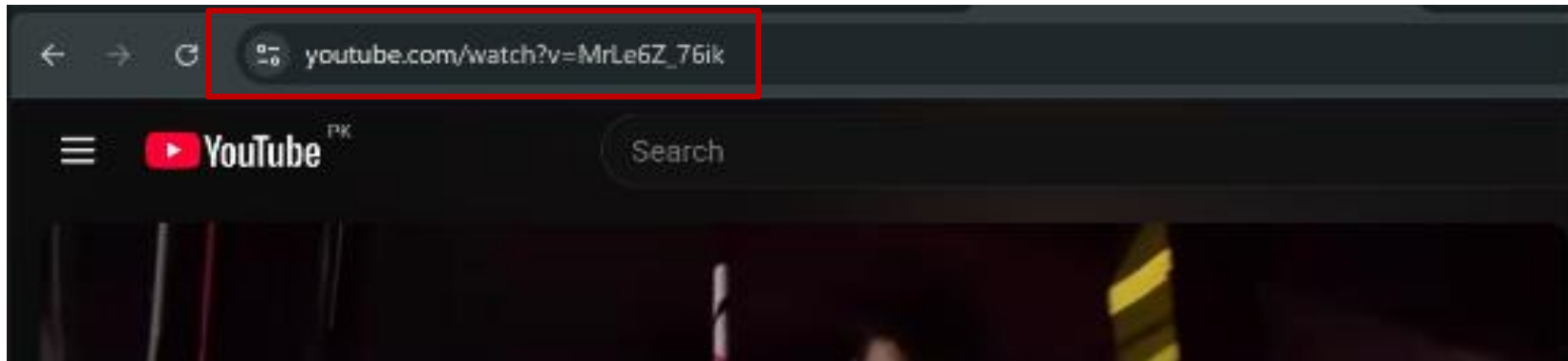


Result [3/10]:

Monitoring Modules On Raspberry Pi 5



Result [4/10]:





Result [5/10]:

Dataset Generation and Tested Ai Classifier on Google Colab

Pi Parental Control – Data Collector

Current Label: **SHORTS**

CSV Total Rows: 0 | Session Total: 0

Session Stats:

- Shorts: 0
- Reels: 0
- TikTok: 0
- Normal: 0
- YouTube Normal: 0

Change Label:

SHORTS

REELS

TIKTOK

NORMAL

YOUTUBE NORMAL

Minimum 200 rows required per label for good ML training.

Refresh this page to update stats.

=====

DATASET INFO

=====

Total Rows : 1312

Total Columns : 32

Label Distribution:

label

normal 378

youtube_normal 250

shorts 249

reels 228

tiktok 207

Name: count, dtype: int64

Binary Labels:

BLOCK (1) - shorts/reels/tiktok : 684

ALLOW (0) - normal browsing : 628

Train set : 1049 rows

Test set : 263 rows

Result [6/10]:



Dataset Generation and Tested Ai Classifier on Google Colab

```
... =====
TRAINING CLASSIFIERS
=====

Training: Logistic Regression...
  Accuracy : 88.21%
  CV Score : 87.80% ± 1.48%
  AUC      : 89.48%
  F1 Score : 88.22%
  Precision : 88.24%
  Recall   : 88.21%

Training: SVM (Linear)...
  Accuracy : 86.69%
  CV Score : 88.66% ± 1.59%
  AUC      : 91.19%
  F1 Score : 86.70%
  Precision : 86.83%
  Recall   : 86.69%
```

```
... Training: SVM (RBF)...
  Accuracy : 88.21%
  CV Score : 90.37% ± 1.54%
  AUC      : 92.73%
  F1 Score : 88.22%
  Precision : 88.29%
  Recall   : 88.21%

Training: Decision Tree...
  Accuracy : 87.83%
  CV Score : 89.14% ± 2.96%
  AUC      : 87.93%
  F1 Score : 87.83%
  Precision : 88.09%
  Recall   : 87.83%
```

```
... Training: Random Forest...
  Accuracy : 88.59%
  CV Score : 90.66% ± 1.36%
  AUC      : 92.89%
  F1 Score : 88.59%
  Precision : 88.59%
  Recall   : 88.59%

Training: Gradient Boosting...
  Accuracy : 88.59%
  CV Score : 90.66% ± 1.18%
  AUC      : 92.01%
  F1 Score : 88.60%
  Precision : 88.61%
  Recall   : 88.59%
```

```
... Training: XGBoost...
  Accuracy : 88.21%
  CV Score : 90.75% ± 1.06%
  AUC      : 93.26%
  F1 Score : 88.22%
  Precision : 88.24%
  Recall   : 88.21%

Training: CatBoost...
  Accuracy : 89.35%
  CV Score : 91.61% ± 0.87%
  AUC      : 93.06%
  F1 Score : 89.36%
  Precision : 89.37%
  Recall   : 89.35%
```



Result [7/10]:

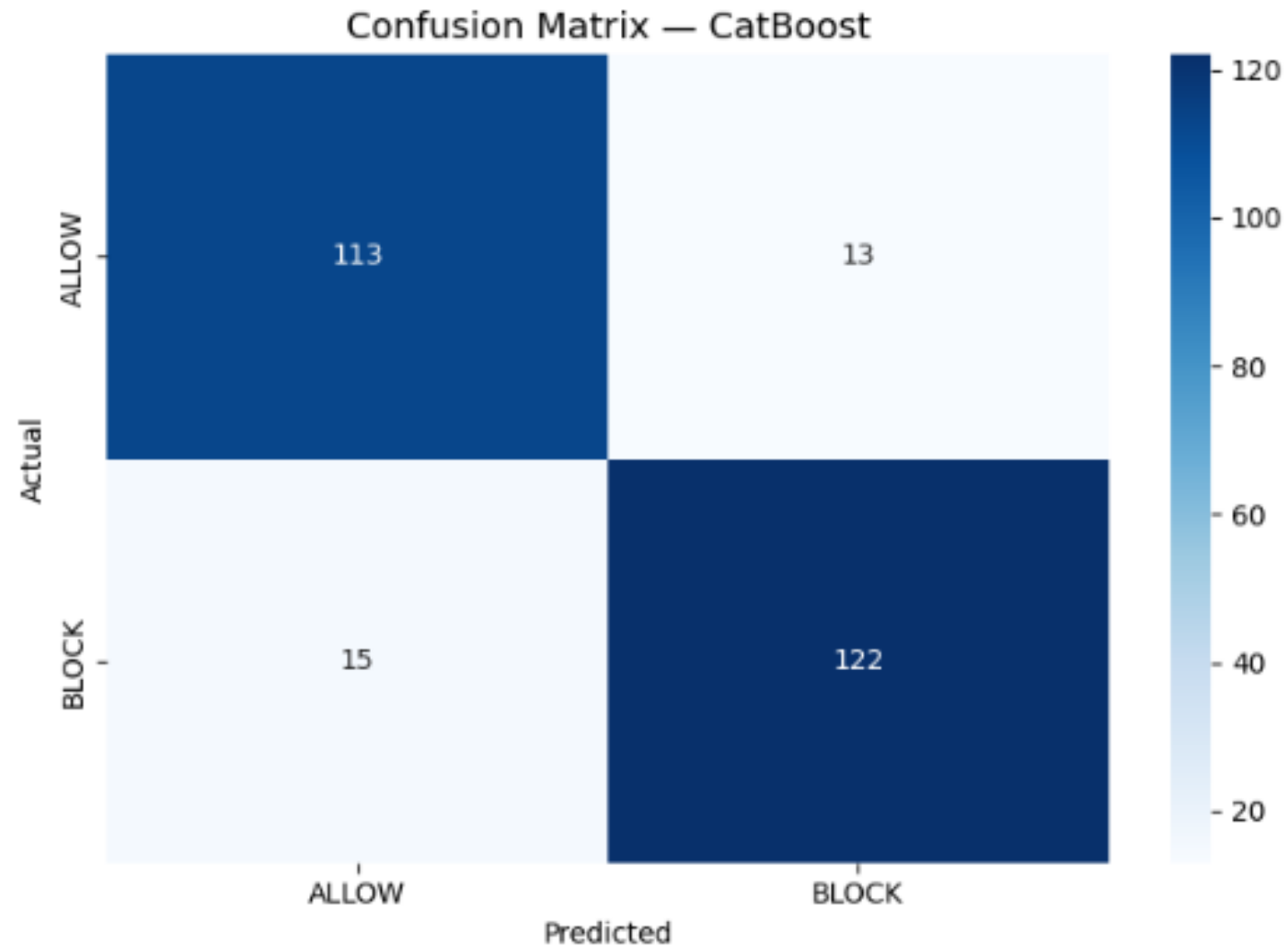
Dataset Generation and Tested Ai Classifier on Google Colab

=====			
Classifier	Accuracy	CV Score	AUC

1 CatBoost	89.35%	91.61%±0.87	93.06%
2 Random Forest	88.59%	90.66%±1.36	92.89%
3 Gradient Boosting	88.59%	90.66%±1.18	92.01%
Logistic Regression	88.21%	87.80%±1.48	89.48%
SVM (RBF)	88.21%	90.37%±1.54	92.73%
XGBoost	88.21%	90.75%±1.06	93.26%
Decision Tree	87.83%	89.14%±2.96	87.93%
SVM (Linear)	86.69%	88.66%±1.59	91.19%
=====			

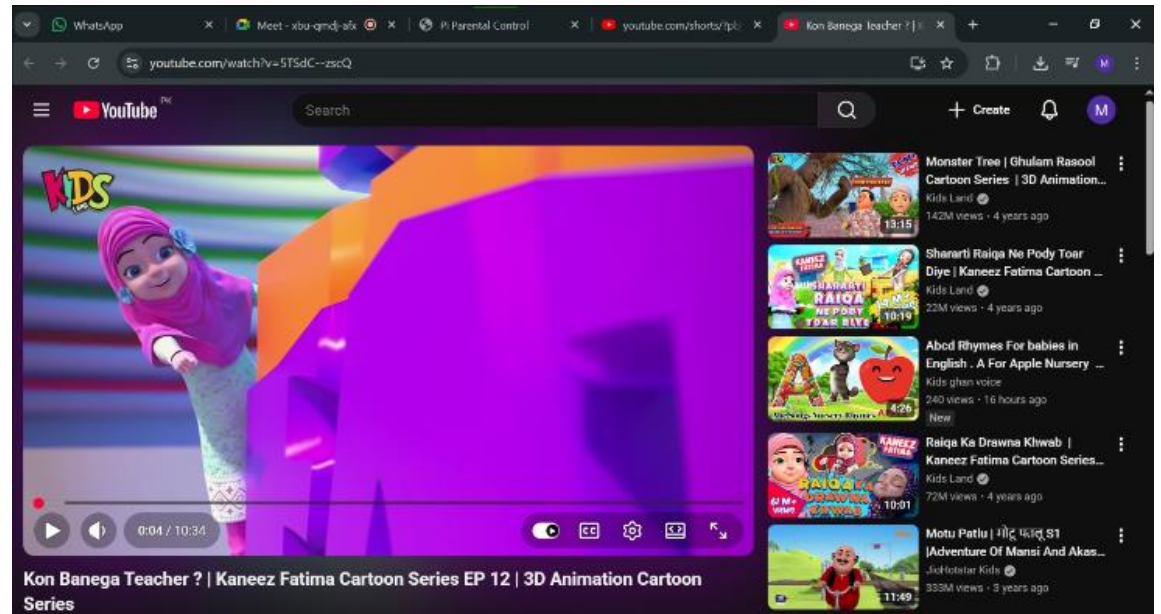
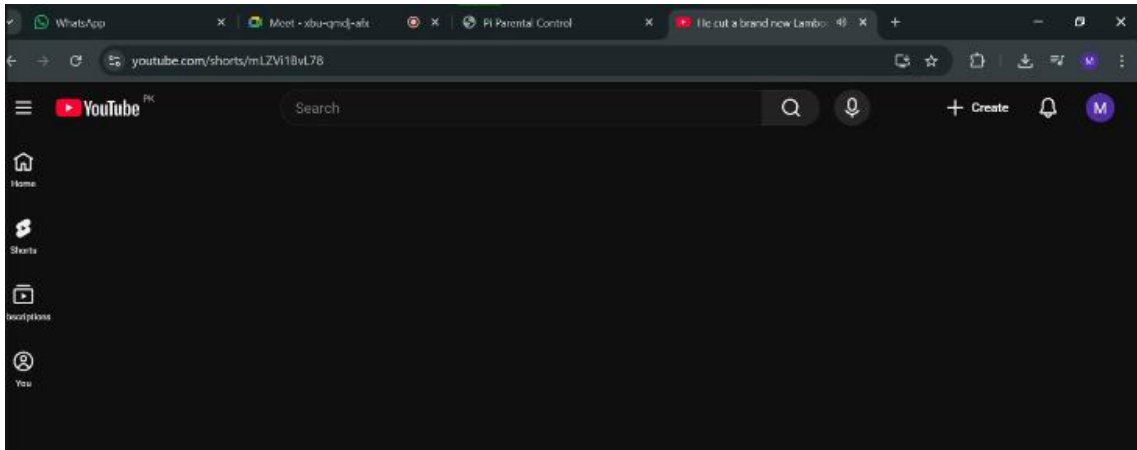
Result [7/10]:

Confusion Matrix For CatBoost



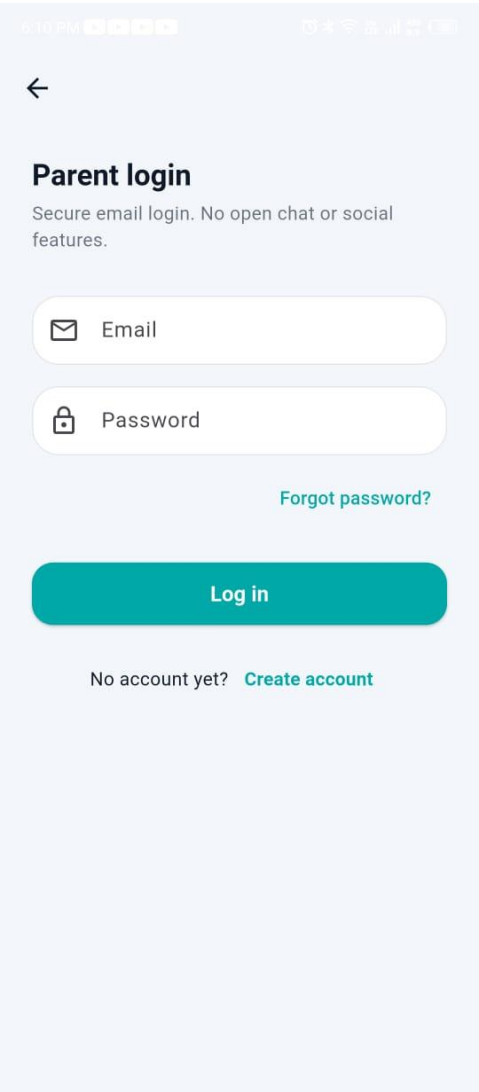
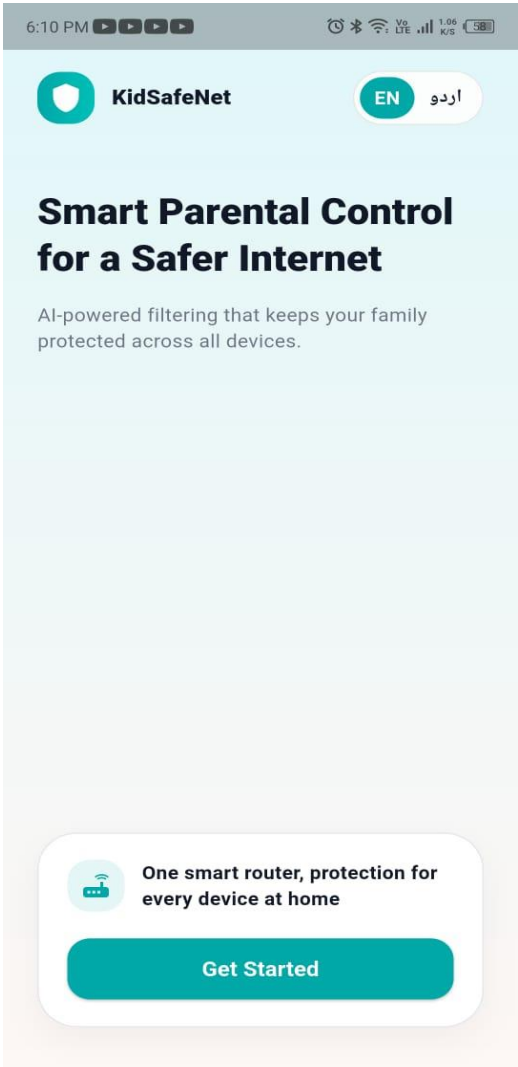
Result [8/10]:

Shorts Blocking and Redirection



Result Module 3 [9/10]:

Mobile APP





Result Module 3 [10/10]:

